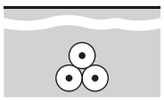
POWER CABLES / IEC 60502-1

CU / PVC / PVC

0.6 / 1 kV

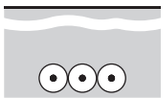
Nominal cross sectional area	ELECTRICAL DATA									DIMENSIONS AND WEIGHTS		Cable Code
	Max. Conductor Resistance		Continuous Current Ratings							Approx. overall diameter	Approx. overall weight	
	DC at 20 °C	AC at 70 °C	Buried direct in ground		In buried ducts		In free air					
			(a)	(b)	(c)	(d)	(e)	(f)	(g)			
mm²	Ω / km	Ω / km	A	A	A	A	A	A	A	mm	kg / km	
1.5	12.1000	14.4777	22	22	16	18	17	17	23	6.0	55	C208PA10100CB51IMR
2.5	7.4100	8.8661	29	29	21	24	23	23	30	6.4	70	C210PA10100CB51IMR
4	4.6100	5.5159	37	37	28	31	30	31	40	7.4	100	C212PA10100CB51IMR
6	3.0800	3.6853	46	46	34	39	38	39	50	7.9	125	C213PA10100CB51IMR
10	1.8300	2.1897	61	61	46	51	52	53	68	8.5	160	C314PA10100CB51IMR
16	1.1500	1.3762	79	79	60	66	69	70	90	9.5	225	C315PA10100CB51IMR
25	0.7270	0.8702	101	101	78	86	92	94	119	11.1	330	C316PA10100CB51IMR
35	0.5240	0.6274	121	121	94	104	113	116	147	12.1	425	C317PA10100CB51IMR
50	0.3870	0.4637	143	143	114	124	138	142	178	13.7	560	C318PA10100CB51IMR
70	0.2680	0.3217	175	175	141	152	175	180	225	15.3	770	C319PA10100CB51IMR
95	0.1930	0.2324	209	209	172	185	216	223	276	17.5	1045	C345PA10100CB51IMR
120	0.1530	0.1850	236	237	197	211	251	258	320	18.8	1285	C346PA10100CB51IMR
150	0.1240	0.1508	265	266	224	238	289	298	367	20.9	1580	C347PA10100CB51IMR
185	0.0991	0.1216	300	300	256	272	335	345	423	23.2	1970	C348PA10100CB51IMR
240	0.0754	0.0942	346	346	300	318	398	410	502	26.1	2555	C349PA10100CB51IMR
300	0.0601	0.0769	389	389	341	362	460	474	581	29.1	3165	C350PA10100CB51IMR
400	0.0470	0.0625	437	437	389	409	532	548	674	32.3	4050	C351PA10100CB51IMF
500	0.0366	0.0515	490	490	442	473	614	632	785	36.3	5190	C352PA10100CB51IMF
630	0.0283	0.0433	544	548	496	537	700	722	910	40.1	6520	C353PA10100CB51IMF
800	0.0221	0.0374	596	596	549	600	787	812	1042	44.0	8310	C354PA10100CB51IMF
1000	0.0176	0.0333	652	655	608	680	909	941	1229	52.8	10465	C255PA10100CB51IMF

Fig. (a)



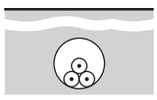
Trefoil

Fig. (b)



Flat touched

Fig. (c)



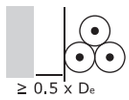
Trefoil

Fig. (d)



Flat touched

Fig. (e)



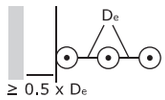
Trefoil

Fig. (f)



Flat touched

Fig. (g)



Flat spaced

Laying Conditions: *Ambient air temperature of* **40 ºC**, *Ambient ground temperature of* **35 ºC**, *Soil thermal resistivity of* **1.5 K·m/W** *and Depth of laying of* **0.5 m**. *In case of different laying conditions, appropriate correction (derating) factors from* ***Annex C*** *have to be applied to cater for the actual installation conditions.*

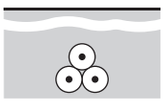
POWER CABLES / IEC 60502-1

CU / XLPE / PVC

0.6 / 1 kV

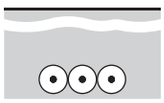
Nominal cross sectional area	ELECTRICAL DATA									DIMENSIONS AND WEIGHTS		Cable Code
	Max. Conductor Resistance		Continuous Current Ratings							Approx. overall diameter	Approx. overall weight	
	DC at 20 °C	AC at 90 °C	Buried direct in ground		In buried ducts		In free air					
mm²	Ω / km	Ω / km	(a)	(b)	(c)	(d)	(e)	(f)	(g)			
			A	A	A	A	A	A	A	mm	kg / km	
1.5	12.1000	15.4287	27	28	20	23	23	23	31	5.8	50	C208XA10100MB51IMR
2.5	7.4100	9.4485	36	36	26	30	30	31	40	6.2	65	C210XA10100MB51IMR
4	4.6100	5.8782	46	46	34	38	40	41	53	6.8	85	C212XA10100MB51IMR
6	3.0800	3.9274	57	57	42	48	50	52	68	7.3	105	C213XA10100MB51IMR
10	1.8300	2.3335	75	75	56	63	68	70	91	7.9	140	C314XA10100MB51IMR
16	1.1500	1.4665	97	97	73	82	91	93	121	8.9	205	C315XA10100MB51IMR
25	0.7270	0.9273	124	124	96	106	122	125	161	10.5	300	C316XA10100MB51IMR
35	0.5240	0.6686	149	149	117	128	150	154	198	11.5	395	C317XA10100MB51IMR
50	0.3870	0.4941	175	176	140	153	183	189	240	12.9	520	C318XA10100MB51IMR
70	0.2680	0.3428	214	215	174	188	233	240	304	14.7	725	C319XA10100MB51IMR
95	0.1930	0.2476	257	256	212	227	288	297	374	16.5	980	C345XA10100MB51IMR
120	0.1530	0.1970	291	292	243	260	335	346	434	18.0	1220	C346XA10100MB51IMR
150	0.1240	0.1605	327	327	277	295	388	400	499	20.1	1500	C347XA10100MB51IMR
185	0.0991	0.1294	369	369	316	336	450	464	577	22.2	1860	C348XA10100MB51IMR
240	0.0754	0.1002	425	426	371	393	536	553	688	24.9	2415	C349XA10100MB51IMR
300	0.0601	0.0817	479	473	422	447	620	641	797	27.7	2990	C350XA10100MB51IMR
400	0.0470	0.0663	539	540	482	512	720	743	928	30.9	3845	C351XA10100MB51IMR
500	0.0366	0.0545	604	606	549	587	834	862	1084	34.9	4950	C352XA10100MB51IMF
630	0.0283	0.0454	674	673	619	668	955	986	1257	39.3	6295	C353XA10100MB51IMF
800	0.0221	0.0390	739	741	688	750	1078	1116	1440	43.6	8090	C354XA10100MB51IMF
1000	0.0176	0.0346	807	808	766	841	1234	1276	1674	52.2	10150	C255XA10100MB51IMF

Fig. (a)



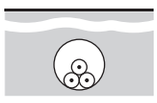
Trefoil

Fig. (b)



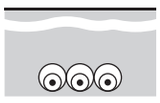
Flat touched

Fig. (c)



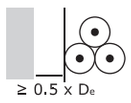
Trefoil

Fig. (d)



Flat touched

Fig. (e)



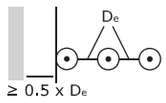
Trefoil

Fig. (f)



Flat touched

Fig. (g)



Flat spaced

Laying Conditions: *Ambient air temperature of* **40 ºC**, *Ambient ground temperature of* **35 ºC**, *Soil thermal resistivity of* **1.5 K·m/W** *and Depth of laying of* **0.5 m**. *In case of different laying conditions, appropriate correction (derating) factors from* ***Annex C*** *have to be applied to cater for the actual installation conditions.*

POWER CABLES / IEC 60502-1

CU / PVC / PVC

0.6 / 1 kV

Nominal cross sectional area		ELECTRICAL DATA					DIMENSIONS AND WEIGHTS		Cable Code
		Max. Conductor Resistance		Continuous Current Ratings			Approx. overall diameter	Approx. overall weight	
		DC at 20 °C	AC at 70 °C	Laid in ground	Laid in ducts	Laid in free air			
mm²		Ω / km	Ω / km	A	A	A	mm	kg / km	
Two Core Cables									
1.5	rm	12.1000	14.4777	27	19	21	9.9	145	C208PA10200CB01IMR
2.5	rm	7.4100	8.8661	36	24	28	10.8	185	C210PA10200CB01IMR
4	rm	4.6100	5.5159	47	32	38	12.7	260	C212PA10200CB01IMR
6	rm	3.0800	3.6853	58	40	48	13.8	330	C213PA10200CB01IMR
10	rm	1.8300	2.1898	76	53	65	15.0	420	C314PA10200CB01IMR
16	rm	1.1500	1.3763	98	70	86	17.0	580	C315PA10200CB01IMR
25	rm	0.7270	0.8703	127	92	116	20.1	855	C316PA10200CB01IMR
35	rm	0.5240	0.6276	152	111	142	22.2	1100	C317PA10200CB01IMR
Three Core Cables									
1.5	rm	12.1000	14.4777	23	15	18	10.4	170	C208PA10300CB04IMR
2.5	rm	7.4100	8.8661	29	20	23	11.4	220	C210PA10300CB04IMR
4	rm	4.6100	5.5159	38	26	32	13.4	315	C212PA10300CB04IMR
6	rm	3.0800	3.6853	48	33	40	14.7	400	C213PA10300CB04IMR
10	rm	1.8300	2.1898	63	43	55	16.4	500	C314PA10300CB04IMR
16	rm	1.1500	1.3763	81	57	72	18.6	720	C315PA10300CB04IMR
25	rm	0.7270	0.8703	105	75	98	21.9	1020	C316PA10300CB04IMR
35	sm	0.5240	0.6276	117	85	107	21.2	1310	C417PA10300CB04IMR
50	sm	0.3870	0.4639	139	103	131	24.6	1715	C418PA10300CB04IMR
70	sm	0.2680	0.3220	170	128	166	27.7	2355	C419PA10300CB04IMR
95	sm	0.1930	0.2328	204	156	204	32.1	3205	C445PA10300CB04IMR
120	sm	0.1530	0.1856	231	178	235	34.0	3940	C446PA10300CB04IMF
Four Core Cables									
1.5	rm	12.1000	14.4777	23	15	18	11.3	205	C208PA10400CB08IMR
2.5	rm	7.4100	8.8661	29	20	23	12.3	265	C210PA10400CB08IMR
4	rm	4.6100	5.5159	38	26	32	14.6	385	C212PA10400CB08IMR
6	rm	3.0800	3.6853	48	33	40	16.0	500	C213PA10400CB08IMR
10	rm	1.8300	2.1898	63	43	55	17.9	670	C314PA10400CB08IMR
16	rm	1.1500	1.3763	81	57	72	20.3	920	C315PA10400CB08IMR
25	rm	0.7270	0.8703	105	75	98	24.1	1295	C316PA10400CB08IMR
35	sm	0.5240	0.6276	117	85	107	24.5	1665	C417PA10400CB08IMR
50	sm	0.3870	0.4639	139	103	131	28.7	2220	C418PA10400CB08IMR
70	sm	0.2680	0.3220	170	128	166	32.3	3055	C419PA10400CB08IMR
95	sm	0.1930	0.2328	204	156	204	36.2	4200	C445PA10400CB08IMR
120	sm	0.1530	0.1856	231	178	235	39.6	5185	C446PA10400CB08IMF
150	sm	0.1240	0.1514	259	203	270	44.3	6390	C447PA10400CB08IMF
185	sm	0.0991	0.1224	292	232	311	49.1	7905	C448PA10400CB08IMF
240	sm	0.0754	0.0952	338	272	370	55.6	10280	C449PA10400CB08IMS
300	sm	0.0601	0.0780	381	310	426	61.7	12840	C450PA10400CB08IMS
400	sm	0.0470	0.0638	431	357	495	70.3	16455	C451PA10400CB08IMS
500	sm	0.0366	0.0531	484	405	568	77.8	21170	C452PA10400CB08IMS
Four Core Cables with Reduced Neutral									
25 rm	16 rm	0.7270 / 1.1500	0.8703 / 1.3763	105	75	98	23.1	1215	C334PA10400CB08IMR
35 sm	16 rm	0.5240 / 1.1500	0.6276 / 1.3763	117	85	107	24.5	1480	C435PA10400CB08IMR
50 sm	25 rm	0.3870 / 0.7270	0.4639 / 0.8703	139	103	131	28.7	2005	C436PA10400CB08IMR
70 sm	35 sm	0.2680 / 0.5240	0.3220 / 0.6276	170	128	166	32.3	2745	C437PA10400CB08IMR
95 sm	50 sm	0.1930 / 0.3870	0.2328 / 0.4639	204	156	204	36.2	3700	C438PA10400CB08IMR
120 sm	70 sm	0.1530 / 0.2680	0.1856 / 0.3220	231	178	235	39.6	4665	C439PA10400CB08IMF
150 sm	70 sm	0.1240 / 0.2680	0.1514 / 0.3220	259	203	270	44.3	5605	C440PA10400CB08IMF
185 sm	95 sm	0.0991 / 0.1930	0.1224 / 0.2328	292	232	311	49.1	7035	C441PA10400CB08IMF
240 sm	120 sm	0.0754 / 0.1530	0.0952 / 0.1856	338	272	370	55.6	9045	C442PA10400CB08IMS
300 sm	150 sm	0.0601 / 0.1240	0.0780 / 0.1514	381	310	426	61.7	11290	C443PA10400CB08IMS
400 sm	185 sm	0.0470 / 0.0991	0.0638 / 0.1224	431	357	495	70.3	14375	C444PA10400CB08IMS
500 sm	240 sm	0.0366 / 0.0754	0.0531 / 0.0952	484	405	568	77.8	18415	C466PA10400CB08IMS

Laying Conditions: *Ambient air temperature of* **40 °C**, *Ambient ground temperature of* **35 °C**, *Soil thermal resistivity of* **1.5 K·m/W** *and Depth of laying of* **0.5 m**. *In case of different laying conditions, appropriate correction (derating) factors from* ***Annex C*** *have to be applied to cater for the actual installation conditions.*

POWER CABLES / IEC 60502-1

CU / XLPE / PVC

0.6 / 1 kV

Nominal cross sectional area		ELECTRICAL DATA					DIMENSIONS AND WEIGHTS		Cable Code
		Max. Conductor Resistance		Continuous Current Ratings			Approx. overall diameter	Approx. overall weight	
		DC at 20 °C	AC at 90 °C	Laid in ground	Laid in ducts	Laid in free air			
mm²		Ω / km	Ω / km	A	A	A	mm	kg / km	
Two Core Cables									
1.5	rm	12.1000	15.4287	33	22	28	9.5	130	C208XA10200MB01IMR
2.5	rm	7.4100	9.4485	43	29	36	10.4	165	C210XA10200MB01IMR
4	rm	4.6100	5.8783	56	38	48	11.5	215	C212XA10200MB01IMR
6	rm	3.0800	3.9274	69	48	61	12.6	275	C213XA10200MB01IMR
10	rm	1.8300	2.3336	91	63	82	13.8	365	C314XA10200MB01IMR
16	rm	1.1500	1.4667	118	83	110	15.8	520	C315XA10200MB01IMR
25	rm	0.7270	0.9275	153	110	149	18.9	775	C316XA10200MB01IMR
35	rm	0.5240	0.6688	183	133	183	21.0	1010	C317XA10200MB01IMR
Three Core Cables									
1.5	rm	12.1000	15.4287	27	18	23	10.1	150	C208XA10300MB04IMR
2.5	rm	7.4100	9.4485	35	24	30	11.0	195	C210XA10300MB04IMR
4	rm	4.6100	5.8783	46	31	40	12.1	260	C212XA10300MB04IMR
6	rm	3.0800	3.9274	57	39	51	13.4	340	C213XA10300MB04IMR
10	rm	1.8300	2.3336	75	53	70	15.1	445	C314XA10300MB04IMR
16	rm	1.1500	1.4667	97	69	93	17.3	660	C315XA10300MB04IMR
25	rm	0.7270	0.9275	126	92	126	20.6	940	C316XA10300MB04IMR
35	sm	0.5240	0.6688	144	105	141	20.0	1210	C417XA10300MB04IMR
50	sm	0.3870	0.4944	171	127	173	23.0	1580	C418XA10300MB04IMR
70	sm	0.2680	0.3431	209	159	220	26.5	2210	C419XA10300MB04IMR
95	sm	0.1930	0.2481	251	193	271	29.9	2970	C445XA10300MB04IMR
120	sm	0.1530	0.1976	286	223	317	33.2	3710	C446XA10300MB04IMF
Four Core Cables									
1.5	rm	12.1000	15.4287	27	18	23	10.8	175	C208XA10400MB08IMR
2.5	rm	7.4100	9.4485	35	24	30	11.9	235	C210XA10400MB08IMR
4	rm	4.6100	5.8783	46	31	40	13.2	315	C212XA10400MB08IMR
6	rm	3.0800	3.9274	57	39	51	14.5	420	C213XA10400MB08IMR
10	rm	1.8300	2.3336	75	53	70	16.4	600	C314XA10400MB08IMR
16	rm	1.1500	1.4667	97	69	93	18.9	840	C315XA10400MB08IMR
25	rm	0.7270	0.9275	126	92	126	22.6	1190	C316XA10400MB08IMR
35	sm	0.5240	0.6688	144	105	141	23.0	1535	C417XA10400MB08IMR
50	sm	0.3870	0.4944	171	127	173	26.5	2020	C418XA10400MB08IMR
70	sm	0.2680	0.3431	209	159	220	30.8	2865	C419XA10400MB08IMR
95	sm	0.1930	0.2481	251	193	271	33.5	3890	C445XA10400MB08IMR
120	sm	0.1530	0.1976	286	223	317	37.6	4890	C446XA10400MB08IMF
150	sm	0.1240	0.1612	319	252	360	42.1	6010	C447XA10400MB08IMF
185	sm	0.0991	0.1302	361	288	416	47.1	7475	C448XA10400MB08IMF
240	sm	0.0754	0.1012	417	339	495	52.9	9685	C449XA10400MB08IMS
300	sm	0.0601	0.0829	470	386	570	58.5	12100	C450XA10400MB08IMS
400	sm	0.0470	0.0676	533	444	665	67.1	15550	C451XA10400MB08IMS
500	sm	0.0366	0.0561	598	505	764	74.6	20100	C452XA10400MB08IMS
Four Core Cables with Reduced Neutral									
25 rm	16 rm	0.7270 / 1.1500	0.9275 / 1.4667	126	92	126	21.7	1115	C334XA10400MB08IMR
35 sm	16 rm	0.5240 / 1.1500	0.6688 / 1.4667	144	105	141	23.0	1365	C435XA10400MB08IMR
50 sm	25 rm	0.3870 / 0.7270	0.4944 / 0.9275	171	127	173	26.5	1825	C436XA10400MB08IMR
70 sm	35 sm	0.2680 / 0.5240	0.3431 / 0.6688	209	159	220	29.2	2545	C437XA10400MB08IMR
95 sm	50 sm	0.1930 / 0.3870	0.2481 / 0.4944	251	193	271	33.0	3440	C438XA10400MB08IMR
120 sm	70 sm	0.1530 / 0.2680	0.1976 / 0.3431	286	223	317	35.9	4380	C439XA10400MB08IMF
150 sm	70 sm	0.1240 / 0.2680	0.1612 / 0.3431	319	252	360	39.7	5260	C440XA10400MB08IMF
185 sm	95 sm	0.0991 / 0.1930	0.1302 / 0.2481	361	288	416	44.6	6630	C441XA10400MB08IMF
240 sm	120 sm	0.0754 / 0.1530	0.1012 / 0.1976	417	339	495	49.9	8525	C442XA10400MB08IMS
300 sm	150 sm	0.0601 / 0.1240	0.0829 / 0.1612	470	386	570	55.2	10630	C443XA10400MB08IMS
400 sm	185 sm	0.0470 / 0.0991	0.0676 / 0.1302	533	444	665	63.1	13545	C444XA10400MB08IMS
500 sm	240 sm	0.0366 / 0.0754	0.0561 / 0.1012	598	505	764	70.3	17465	C466XA10400MB08IMS

Laying Conditions: *Ambient air temperature of* **40 ºC**, *Ambient ground temperature of* **35 ºC**, *Soil thermal resistivity of* **1.5 K·m/W** *and Depth of laying of* **0.5 m**. *In case of different laying conditions, appropriate correction (derating) factors from* ***Annex C*** *have to be applied to cater for the actual installation conditions.*

Annex A: Guides to Use

A.1 Aim

The aim of this annex is to inform users of characteristics and limitations of electric cables and thereby to minimize their misuse.

It is assumed that the design of installation and the specification, purchase and installation of cables is entrusted to suitably skilled and competent people.

In cases of doubt as to the suitability of cables for a particular use further specific information should be obtained from the manufacturer.

A.2 Cable Selection and Design

A.2.1 The products specified in this catalogue are intended to be used for the supply of electrical energy up to rated voltage of 0.6/1 (1.2) kV. This rating should not be exceeded.

A.2.2 These cables are intended for use within a nominal power frequency range of 49 Hz to 61 Hz.

A.2.3 In addition to the current ratings, due regard should be given to:

a. The capability of the cable to withstand the worst anticipated fault condition of the system;
b. The earth loop impedance;
c. The operating characteristics of the connected equipment;
d. The voltage drop requirements during normal load or motor starting conditions.

A.2.4 The possible effects of transient over-voltages should be recognized as they can be detrimental to cables.

A.3 Environment and Application

A.3.1 Reasonable protection against mechanical damage, appropriate to the choice of cable and the installation conditions, should be provided.

A.3.2 Cables included in this catalogue contain PVC, and when exposed to the effects of external fire will produce harmful smoke and fumes, and where bundled together may propagate fire. When circumstances dictate, consideration should be given to the use of any of our Fire Survival Cables products. Please refer to **alfanar**'s Fire Survival Cables catalogue for more information.

A.3.3 Cables can be harmed by exposure to corrosive products or solvent substances, especially petroleum based chemicals or their vapours.

A.3.4 Cables specified in this catalogue are not specifically designed for use:

a. As self-supporting aerial cables;
b. As submarine cable or for laying in water-logged conditions;
c. Where subsidence is likely, unless special precautions are taken to minimize damage;
d. Where any exposure to excessive heat under normal operating conditions is involved:
e. Where the sheath is subjected to a voltage test after installation.

A.3.5 If cables are exposed to localized heat, solar radiation or high temperature ambient conditions or there is a possibility of higher soil thermal resistivity, the current carrying capacity will be reduced.

A.3.6 The standard sheathing compound (PVC) supplied on these cables do not provide protection against damage by rodents, termites etc., unless requested earlier at time of purchase order.

A.3.7 loaded cables can have surface temperatures that require protection against accidental contact.

Annex A: Guides to Use

A.4 Installation (Please refer to Annex I for more information)

A.4.1 Precautions should be taken to avoid mechanical damage to the cables before and during installation.

A.4.2 Exceeding the manufacturer's recommended maximum pulling tensions can result in damage to the cable.

A.4.3 If cables are to be installed in ducts, the correct size of duct should be used.

A.4.4 The type of jointing and filling compounds employed should be chemically compatible with the cable materials.

A.4.5 The cable support system should be such as to avoid damage or danger under normal or fault conditions.

A.4.6 Cables specified in this catalogue are designed for fixed installations only; for example they are not for use as trailing or reeling cables.

A.4.7 Repeated over-voltage testing can lead to premature failure of the cable.

A.4.8 The selection of cable glands, accessories and any associated tools should take account of all aspects of intended use.

A.5 Storage and handling of drums (Please refer to Annex I for more information)

A.5.1 Cable drums should be regularly inspected during storage to assess their physical condition.

A.5.2 Battens, where applied, should not be removed from drums until the cable is about to be installed.

A.5.3 When handling drums, reasonable precautions should be taken to avoid injury. Due regard should be paid to the weight, method and direction of rolling, lifting, protruding nails and splinters.

A.5.4 Care should be taken to avoid deterioration of drums or their becoming a hazard to the general public.

A.5.5 A detailed guide to the safe handling of cable drums is available in Annex (I) of this catalogue.

Annex B: Continuous Current Ratings

B.1 General

This annex deals solely with the installation conditions used to calculate the steady-state continuous current ratings of single-core and multi-core cables having extruded insulation. The tabulated current ratings provided in this catalogue have been calculated for cables having a rated voltage of 0.6/1 kV and constructions as detailed in each relevant type.

The tabulated current ratings in this catalogue have been calculated using the methods set out in IEC 60287.

B.2 Temperatures

The maximum conductor temperatures for which the tabulated current ratings have been calculated are 70 °C and 90 °C for PVC and XLPE insulated cables respectively.

The reference ambient temperatures assumed are as follows:

– For cables in free air: **40 °C**

– For buried cables, either directly in the soil or in ducts in the ground: **35 °C**

Derating factors for other ambient temperatures are given in Tables C.1 and C.2.

The current ratings for cables in air do not take account of the increase, if any, due to solar or other infra-red radiation. Where the cables are subject to such radiation, the current rating should be derived by the methods specified in IEC 60287.

B.3 Soil thermal resistivity

The tabulated current ratings in this catalogue for cables in ducts or direct in the ground relate to a soil thermal resistivity of **1.5 °C.m/W**. Derating factors for other values of thermal resistivity are given in tables C.4 to C.7.

It is assumed that the soil properties are uniform; no allowance has been made for the possibility of moisture migration, which can lead to a region of high thermal resistivity around the cable. If partial drying-out of the soil is foreseen, the permissible current rating should be derived by the methods specified in IEC 60287.

B.4 Methods of installation

Current ratings tabulated in this catalogue are for cables installed in the following conditions.

B.4.1 Single-core cables buried direct

Current ratings are given for cables buried direct in the ground at a depth of **0.5 m** (derating factors for other values of burial depth are given in table C.3) under the following conditions:

1. Three cables in trefoil formation touching throughout their length **Fig. (a)**;
2. Three cables in horizontal flat formation touching throughout their length, **Fig. (b)**.

The cable depth is measured to the cable axis or centre of the trefoil group.

B.4.2 Single-core cables in polyvinyl chloride (PVC) ducts

Current ratings are given for cables in polyvinyl chloride (PVC) ducts buried at a depth of 0.5 m (derating factors for other values of burial depth are given in table C.3) with one cable per duct as follows:

1. Three cables in trefoil formation in single duct **Fig. (c)**;
2. Three cables in horizontal flat formation, ducts touching throughout their length **Fig. (d)**.

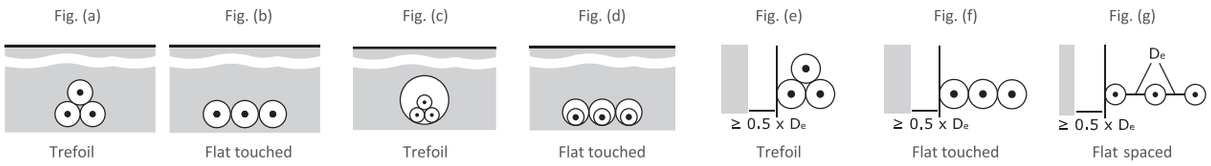
The ducts are assumed to be polyvinyl chloride (PVC) having an inside diameter of 1.5 times the outside diameter of the cable and a wall thickness equal to 6 % of the duct inside diameter. The ratings are based on the assumption that the ducts are air filled. The tabulated ratings may be applied to cables in ducts having an inside diameter of between 1.2 and 2 times the outside diameter of the cables. For this range of diameters, the variation in the rating is less than 2 % of the tabulated value.

Annex B: Continuous Current Ratings

B.4.3 Single-core cables in air

The cables are assumed to be spaced at least 0.5 times the cable diameter D_e from any vertical surface and installed on brackets or ladder racks as follows:

1. Three cables in trefoil formation touching throughout their length **Fig. (e)**.
2. Three cables in horizontal flat formation touching throughout their length **Fig. (f)**.
3. Three cables in horizontal flat formation with a clearance of one cable diameter D_e **Fig. (g)**.



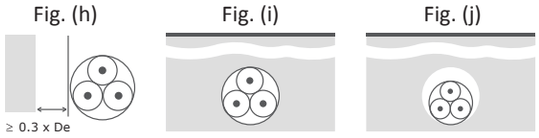
B.4.4 Three-core cables

Current ratings are given for multi-core cables installed under the following conditions:

1. Single cable in air spaced at least 0.3 times the cable diameter D_e from any vertical surface **Fig. (h)**.

2. Single cable buried direct in the ground at a depth of 0.5 m (derating factors for other values of burial depth are given in table C.3) **Fig. (i)**.

3. Single cable in a buried polyvinyl chloride (PVC) duct having dimensions calculated in the same manner as for the single-core cables in ducts. The depth of burial of the duct is 0.5 m (derating factors for other values of burial depth are given in table C.3) **Fig. (j)**.



B.5 Cable loading

The tabulated ratings relate to circuits carrying a balanced three-phase load at a rated frequency of 60 Hz. However, the tabulated ratings can be safely used with circuits carrying a balanced three-phase load at a rated frequency of 50 Hz, where the continuous current rating values are slightly higher in case of rated frequency of 50 Hz.

B.6 Rating factors for grouped circuits

The tabulated current ratings apply to a set of three single-core cables or one multi-core cable forming either a single-phase circuit (two-core cables) or a three-phase circuit. When a number of circuits are installed in close proximity, the rating should be reduced by the appropriate factor from tables C.8 to C.12.

These rating factors should also be applied to groups of parallel cables forming the same circuit. In such cases, attention should also be given to the arrangement of the cables to ensure that the load current is shared equally between the parallel cables.

B.7 Derating factors

The derating factors given in tables C.1 to C.12 for temperature, installation conditions and grouping are averages over a range of conductor sizes and cable types. For particular cases, the derating factor may be calculated using the methods in IEC 60287.

Annex C: Derating Factors

Table C.1

Derating factors for ambient ground temperature

Max. Conductor temperature	Ambient ground temperature °C										
	10	15	20	25	30	35	40	45	50	55	60
70 °C (PVC)	1.31	1.25	1.19	1.13	1.06	1.00	0.92	0.85	0.75	0.65	0.54
90 °C (XLPE)	1.20	1.17	1.12	1.08	1.04	1.00	0.96	0.90	0.85	0.80	0.73

Table C.2

Derating factors for ambient air temperature

Max. Conductor temperature	Ambient air temperature °C										
	10	15	20	25	30	35	40	45	50	55	60
70 °C (PVC)	1.40	1.34	1.29	1.22	1.15	1.08	1.00	0.91	0.82	0.70	0.57
90 °C (XLPE)	1.26	1.23	1.19	1.14	1.10	1.05	1.00	0.96	0.90	0.84	0.78

Table C.3

Derating factors for depth of laying

Depth of laying (mt.)	Cables cross section		
	Up to 70 mm²	95 up to 240 mm²	300 mm² & above
0.50	1.00	1.00	1.00
0.60	0.99	0.98	0.97
0.80	0.97	0.96	0.94
1.00	0.95	0.93	0.92
1.25	0.94	0.92	0.89
1.50	0.93	0.90	0.87
1.75	0.92	0.89	0.86
2.00	0.91	0.88	0.85

Table C.4

Derating factors for soil thermal resistivities for cables buried direct in the ground or in buried ducts

Method of installation	Values of soil thermal resistivity K.m/W						
	0.5	0.7	1.0	1.5	2.0	2.5	3.0
Direct buried cables	1.47	1.27	1.17	1.00	0.88	0.78	0.70
Cable in buried ducts	1.16	1.09	1.07	1.00	0.95	0.91	0.87

NOTE 1 : Values given are averages for the cable types and range of conductor sizes considered in this catalogue. The overall accuracy of derating factors is within ±5 %.

Annex C: Derating Factors

Table C.5

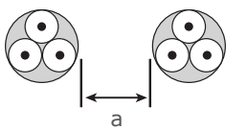
Derating factors for more than one circuit of single-core or multi-core cables laid directly in the ground (Fig. a)

Number of circuits	Circuit to circuit clearance (a)				
	Cables touching	One cable diameter	0.125 m	0.25 m	0.5 m
2	0.75	0.80	0.85	0.90	0.90
3	0.65	0.70	0.75	0.80	0.85
4	0.60	0.60	0.70	0.75	0.80
5	0.55	0.55	0.65	0.70	0.80
6	0.50	0.55	0.60	0.70	0.80
7	0.45	0.51	0.59	0.67	0.76
8	0.43	0.48	0.57	0.65	0.75
9	0.41	0.46	0.55	0.63	0.74
12	0.36	0.42	0.51	0.59	0.71
16	0.32	0.38	0.47	0.56	0.68
20	0.29	0.35	0.44	0.53	0.66

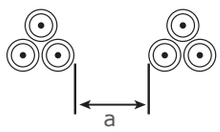
NOTE 1 : Values given are averages for the cable types and range of conductor sizes considered in this catalogue. The process of averaging, together with rounding off, can result in some cases in errors up to ±10 %. Where more precise values are required they may be calculated by methods given in IEC 60287-2-1.

NOTE 2 : If a circuit consists of (m) parallel conductors per phase, then for determining the reduction factor, this circuit should be considered as (m) circuits.

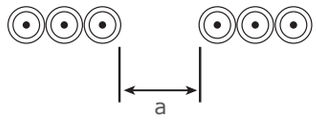
Fig. a



Multi-core cables laid in horizontal formation



Single-core cables laid in trefoil formation



Single-core cables laid in flat formation

Annex C: Derating Factors

Table C.6

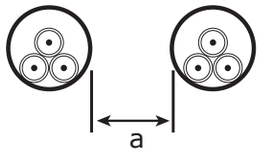
Derating factors for more than one circuit of single-core cables laid in single-way ducts in the ground (Fig. b)

Number of circuits	Duct to duct clearance (a)			
	Ducts touching	0.25 m	0.5 m	1.0 m
2	0.80	0.90	0.90	0.95
3	0.70	0.80	0.85	0.90
4	0.65	0.75	0.80	0.90
5	0.60	0.70	0.80	0.90
6	0.60	0.70	0.80	0.90
7	0.53	0.66	0.76	0.87
8	0.50	0.63	0.74	0.87
9	0.47	0.61	0.73	0.86
10	0.45	0.59	0.72	0.85
11	0.43	0.57	0.70	0.85
12	0.41	0.56	0.69	0.84
13	0.39	0.54	0.68	0.84
14	0.37	0.53	0.68	0.83
15	0.35	0.52	0.67	0.83
16	0.34	0.51	0.66	0.83
17	0.33	0.50	0.65	0.82
18	0.31	0.49	0.65	0.82
19	0.30	0.48	0.64	0.82
20	0.29	0.47	0.63	0.81

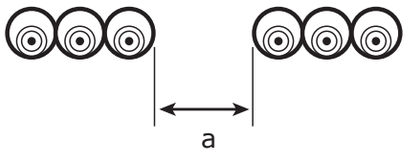
NOTE 1 : Values given are averages for the cable types and range of conductor sizes considered in this catalogue. The process of averaging, together with rounding off, can result in some cases in errors up to ±10 %. Where more precise values are required they may be calculated by methods given in IEC 60287-2-1.

NOTE 2 : If a circuit consists of (n) parallel conductors per phase, then for determining the reduction factor, this circuit should be considered as (n) circuits.

Fig. b



Single-core cables laid in trefoil formation in duct



Single-core cables in single-way ducts laid in flat formation

Annex C: Derating Factors

Table C.7

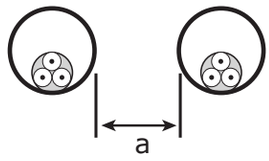
Derating factors for more than one circuit of multi-core cables laid in single-way ducts in the ground (Fig. c)

Number of circuits	Duct to duct clearance (a)			
	Ducts touching	0.25 m	0.5 m	1.0 m
2	0.85	0.90	0.95	0.95
3	0.75	0.85	0.90	0.95
4	0.70	0.80	0.85	0.90
5	0.65	0.80	0.85	0.90
6	0.60	0.80	0.80	0.90
7	0.57	0.76	0.80	0.88
8	0.54	0.74	0.78	0.88
9	0.52	0.73	0.77	0.87
10	0.49	0.72	0.76	0.86
11	0.47	0.70	0.75	0.86
12	0.45	0.69	0.74	0.85
13	0.44	0.68	0.73	0.85
14	0.42	0.68	0.72	0.84
15	0.41	0.67	0.72	0.84
16	0.39	0.66	0.71	0.83
17	0.38	0.65	0.70	0.83
18	0.37	0.65	0.70	0.83
19	0.35	0.64	0.69	0.82
20	0.34	0.63	0.68	0.82

NOTE 1 : Values given are averages for the cable types and range of conductor sizes considered in this catalogue. The process of averaging, together with rounding off, can result in some cases in errors up to ±10 %. Where more precise values are required they may be calculated by methods given in IEC 60287-2-1.

NOTE 2 : If a circuit consists of (n) parallel conductors per phase, then for determining the reduction factor, this circuit should be considered as (n) circuits.

Fig. c

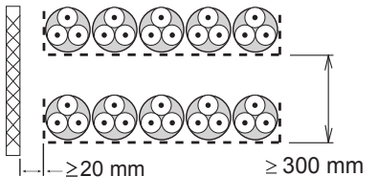
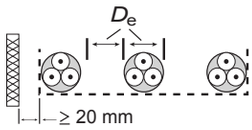


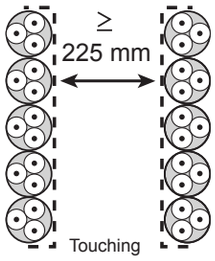
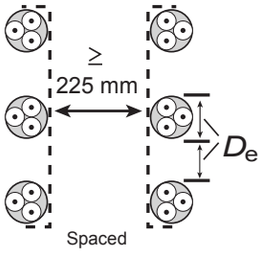
ulti-core cables in single-way duc
laid in horizontal formation

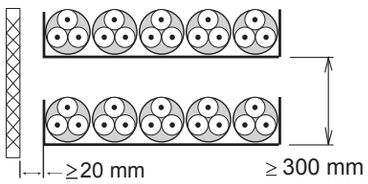
Annex C: Derating Factors

Table C.8

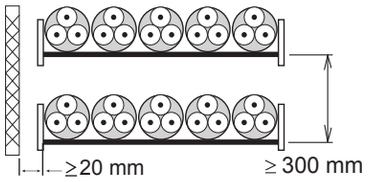
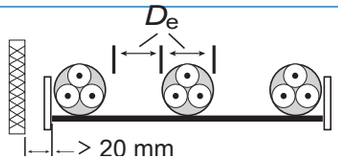
Derating factors for group of more than one multi-core cable in free air

Method of Installation		No. of trays	No. of cables per tray					
			1	2	3	4	6	9
Cables on horizontal perforated cable tray systems (note 4)	 Touching	1	1.00	0.88	0.82	0.79	0.76	0.73
		2	1.00	0.87	0.80	0.77	0.73	0.68
		3	1.00	0.86	0.79	0.76	0.71	0.66
		6	1.00	0.84	0.77	0.73	0.68	0.64
	 Spaced	1	1.00	1.00	0.98	0.95	0.91	-
		2	1.00	0.99	0.96	0.92	0.87	-
		3	1.00	0.98	0.95	0.91	0.85	-

Method of Installation		No. of trays	No. of cables per tray					
			1	2	3	4	6	9
Cables on vertical perforated cable tray systems (note 5)	 Touching	1	1.00	0.88	0.82	0.78	0.73	0.72
		2	1.00	0.88	0.81	0.76	0.71	0.70
	 Spaced	1	1.00	0.91	0.89	0.88	0.87	-
		2	1.00	0.91	0.88	0.87	0.85	-

Method of Installation		No. of trays	No. of cables per tray					
			1	2	3	4	6	9
Cables on horizontal unperforated cable tray systems	 Touching	1	0.97	0.84	0.78	0.75	0.71	0.68
		2	0.97	0.83	0.76	0.72	0.68	0.63
		3	0.97	0.82	0.75	0.71	0.66	0.61
		6	0.97	0.81	0.73	0.69	0.63	0.58

Annex C: Derating Factors

Method of Installation		No. of ladders	No. of cables per ladder					
			1	2	3	4	6	9
Cables on horizontal cable ladder systems, cleats, etc. (note 4)	 Touching	1	1.00	0.87	0.82	0.80	0.79	0.78
		2	1.00	0.86	0.80	0.78	0.76	0.73
		3	1.00	0.85	0.79	0.76	0.73	0.70
		6	1.00	0.84	0.77	0.73	0.68	0.64
	 Spaced	1	1.00	1.00	0.98	0.95	0.91	-
		2	1.00	0.99	0.96	0.92	0.87	-
		3	1.00	0.98	0.95	0.91	0.85	-

NOTE 1 : Values given should be used as a multiplier to current-carrying capacity for one multi-core cable in free air.

NOTE 2 : Values given are averages for the cable types and range of conductor sizes considered in this catalogue. The spread of values is generally less than 5 %.

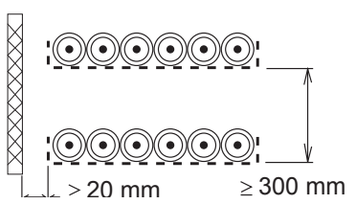
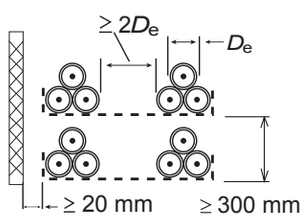
NOTE 3 : Factors apply to single layer groups of cables as shown above and do not apply when cables are installed in more than one layer touching each other. Values for such installations may be significantly lower and must be determined by an appropriate method.

NOTE 4 : Values are given for vertical spacing between trays/ladders of 300 mm and at least 20 mm between trays/ladders and wall. For closer spacing, the factors should be reduced.

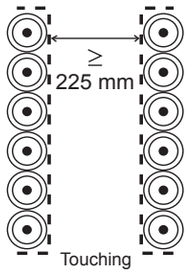
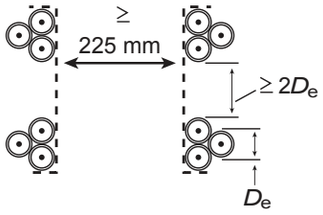
NOTE 5 : Values are given for horizontal spacing between trays of 225 mm with trays mounted back to back. For closer spacing, the factors should be reduced.

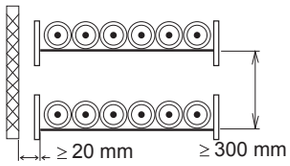
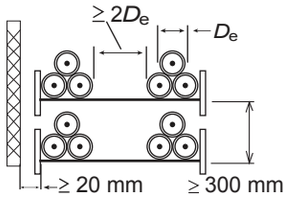
Table C.9

Derating factors for groups of one or more circuits of single-core cables in free air

Method of Installation		No. of trays	No. of three-phase circuits per tray			Use as a multiplier to current-carrying capacity for
			1	2	3	
Cables on horizontal perforated cable tray systems (note 3)	 Touching	1	0.98	0.91	0.87	Three cables in flat formation
		2	0.96	0.87	0.81	
		3	0.95	0.85	0.78	
	 Spaced	1	1.00	0.98	0.96	Three cables in trefoil formation
		2	0.97	0.93	0.89	
		3	0.96	0.92	0.86	

Annex C: Derating Factors

Method of Installation		No. of trays	No. of three-phase circuits per tray			Use as a multiplier to current-carrying capacity for
			1	2	3	
Cables on vertical perforated cable tray systems (note 4)	 Touching	1	0.96	0.86	-	Three cables in flat formation
		2	0.95	0.84	-	
	 Spaced	1	1.00	0.91	0.89	Three cables in trefoil formation
		2	1.00	0.90	0.86	

Method of Installation		No. of ladders	No. of three-phase circuits per ladder			Use as a multiplier to current-carrying capacity for
			1	2	3	
Cables on horizontal cable ladder systems, cleats, etc. (note 3)	 Touching	1	1.00	0.97	0.96	Three cables in flat formation
		2	0.98	0.93	0.89	
		3	0.97	0.90	0.86	
	 Spaced	1	1.00	1.00	1.00	Three cables in trefoil formation
		2	0.97	0.95	0.93	
		3	0.96	0.94	0.90	

NOTE 1 : Values given are averages for the cable types and range of conductor sizes considered in this catalogue. The spread of values is generally less than 5 %.

NOTE 2 : Factors are given for single layers of cables (or trefoil groups) as shown in the table and do not apply when cables are installed in more than one layer touching each other. Values for such installations may be significantly lower and should be determined by an appropriate method.

NOTE 3 : Values are given for vertical spacing between cable trays/ladders of 300 mm and at least 20 mm between cable trays/ladders and wall. For closer spacing the factors should be reduced.

NOTE 4 : Values are given for horizontal spacing between cable trays of 225 mm with cable trays mounted back to back. For closer spacing the factors should be reduced.

NOTE 5 : If a circuit consists of (m) parallel conductors per phase, then for determining the reduction factor this circuit should be considered as (m) circuits.

Annex D: Formulas

D.1 Resistance

The values of conductor DC resistance given in the previous tables are based on 20 °C. In case the DC resistance is required at any other temperature, the following formula is used:

$$R_\theta = R_{20} \times [1+\alpha\ (\theta - 20)] \qquad \Omega/km$$

Where

R_θ	Conductor DC resistance at θ °C	Ω /km
R_{20}	Conductor DC resistance at 20 °C	Ω /km
θ	Operating temperature	°C
α	Resistance temperature coefficient = 0.00393 for Copper = 0.00403 for Aluminum	1 / °C

To get the AC resistance of the conductor at its operating temperature, the following formula is used:

$$R_{a.c} = R_\theta \times (1 + y_p + y_s) \qquad \Omega/km$$

Where

y_p and y_s are the proximity and skin effect factors, respectively, which depend on the laying and operating frequency of the cable.

D.2 Inductance

Self and mutual inductance are formulated as follows:

$$L = K + 0.2\ ln\ \left(\frac{2S}{d}\right) \qquad mH/km$$

Where

L	Inductance	mH/km
K	Constant depends on the conductors' number of wires	
d	Conductor diameter	mm
S	Axial spacing between cables = 1 x S in case of trefoil formation = 1.26 X S in case of flat formation	mm

D.3 Capacitance

$$C = \frac{\epsilon_r}{18\ ln\frac{D}{d}} \qquad \mu F/km$$

Where

C	Capacitance	μF/km
ϵ_r	Relative permittivity of insulation	
D	Diameter over insulation,	mm
d	Diameter under insulation	mm

Annex D: Formulas

D.4 Insulation Resistance

$$R = k \ln \left(\frac{D}{d} \right) \quad M\Omega.km$$

Where

R	Insulation resistance	MΩ.km
K	Constant depends on the insulation	
d	Diameter of the conductor	mm
D	Diameter of the insulated core	mm

D.5 Charging Current

The charging current is the capacitive current which flows when an AC voltage is applied to the cables as a result of the capacitance between the conductor and earth, and for a multi-core cable in which cores are not screened, between conductors. The value can be derived from following the equation:

$$I_c = U_o \omega C 10^{-6} \quad A/km$$

Where

I_c	Charging current	A/km
U_o	Phase voltage	V
ω	$2 \pi f$	
f	Operating frequency	Hz
C	Capacitance to neutral	µF/km

D.6 Dielectric Losses

The dielectric losses of an AC cable are proportional to the capacitance, the frequency, the phase voltage and the power factor. The value can be derived from the following equation:

$$W_d = \omega C U_o^2 \tan \delta 10^{-6} \quad W/km/Ph$$

Where

W_d	Dielectric Losses	W/km/Ph
ω	$2 \pi f$	
f	Operating frequency	Hz
C	Capacitance to neutral	µF/km
U_o	Phase voltage	V
$\tan \delta$	Dielectric power factor	

Annex D: Formulas

D.7 Short Circuit Capacity

Tables F.1 to F.4 give the short circuit current for conductor based on the following conditions:

1. Short circuit starts from the maximum operating conductor temperature of 70 ºC and 90 °C for PVC and XLPE insulation respectively
2. Maximum temperature during short circuit is 160 °C (Cross section ≤ 300 mm^2) and 140 ºC (Cross section > 300 mm^2) for PVC insulation and 250 ºC for XLPE insulation.
3. Maximum short circuit duration is 5 seconds.

If the short circuit current is required at duration not mentioned in the catalogue, it is obtained by dividing the short circuit current for 1 second by the square root of the required duration as follows:

Where

$I_{s.c.t.}$	Short circuit current for t second	kA
$I_{s.c.1}$	Short circuit current for 1 second	kA
t	Duration	Sec.

D.8 Voltage Drop

When current flows in a cable conductor there is a voltage drop between the ends of the conductors, which is the product of the current and the impedance. The following equations should be used to calculate the voltage drop:

A. Single phase circuit:

$$V_d = 2I\ell(R\cos F + X\sin F) \quad V$$

B. Three phase circuit:

$$V_d = \sqrt{3}I\ell(R\cos F + X\sin F) \quad V$$

Where

V_d	Voltage drop	V
I	Load current	A
ℓ	Route length	km
R	AC Resistance	Ω/km
X	Reactance	Ω/km
cos F	Power factor	

Where

$$X = \omega L\ 10^{-3} \quad \Omega/km$$

Where

ω	2 π f	
f	Operating frequency	Hz
L	inductance	mH/km

Relation between cos F and sin F:

cos F	1.0	0.9	0.85	0.8	0.6
sin F	0.0	0.44	0.53	0.6	0.8

- LV cabling system should be planned so as not to exceed voltage drop of 3-5% in normal operating conditions.
- Voltage drop data for LV single and multi core cables are tabulated in table G.1 and G.2.

Annex E: Electrical Parameters Of Cables

Table E.1

Single-core cables with copper conductors - PVC Insulated

Nominal area of conductor	Max. Conductor Resistance		Reactance at (60 Hz, Trefoil)	Impedance at (70 °C, 60 Hz)	Capacitance
	DC at 20 °C	AC at 70 °C			
mm²	Ω / km	Ω / km	Ω / km	Ω / km	µF / km
Non-armoured Cables					
1.5	12.1000	14.4777	0.178	14.479	0.619
2.5	7.4100	8.8661	0.164	8.868	0.763
4	4.6100	5.5159	0.157	5.518	0.753
6	3.0800	3.6853	0.147	3.688	0.904
10	1.8300	2.1897	0.139	2.194	1.028
16	1.1500	1.3762	0.130	1.382	1.254
25	0.7270	0.8702	0.125	0.879	1.271
35	0.5240	0.6274	0.119	0.639	1.489
50	0.3870	0.4637	0.113	0.477	1.497
70	0.2680	0.3217	0.108	0.339	1.753
95	0.1930	0.2324	0.106	0.255	1.782
120	0.1530	0.1850	0.103	0.212	1.964
150	0.1240	0.1508	0.102	0.182	1.955
185	0.0991	0.1216	0.101	0.158	1.969
240	0.0754	0.0942	0.100	0.137	2.042
300	0.0601	0.0769	0.098	0.125	2.113
400	0.0470	0.0625	0.097	0.115	2.189
500	0.0366	0.0515	0.095	0.108	2.318
630	0.0283	0.0433	0.093	0.103	2.580
800	0.0221	0.0374	0.092	0.099	2.899
1000	0.0176	0.0333	0.089	0.095	3.314
Armoured Cables (1)					
6	3.0800	3.6853	0.187	3.690	0.904
10	1.8300	2.1897	0.178	2.197	1.028
16	1.1500	1.3761	0.165	1.386	1.254
25	0.7270	0.8701	0.156	0.884	1.271
35	0.5240	0.6273	0.149	0.645	1.489
50	0.3870	0.4635	0.140	0.484	1.497
70	0.2680	0.3214	0.133	0.348	1.753
95	0.1930	0.2320	0.128	0.265	1.782
120	0.1530	0.1844	0.125	0.223	1.964
150	0.1240	0.1501	0.121	0.193	1.955
185	0.0991	0.1208	0.118	0.169	1.969
240	0.0754	0.0933	0.115	0.148	2.042
300	0.0601	0.0757	0.114	0.137	2.113
400	0.0470	0.0611	0.112	0.127	2.189
500	0.0366	0.0499	0.109	0.120	2.318
630	0.0283	0.0415	0.106	0.114	2.580
800	0.0221	0.0352	0.105	0.111	2.899
1000	0.0176	0.0312	0.101	0.105	3.314

(1) The tabulated values are for armoured cables with round aluminium wires (AWA) only.

Annex E: Electrical Parameters Of Cables

Table E.2

Single-core cables with aluminium conductors - PVC Insulated

Nominal area of conductor	Max. Conductor Resistance		Reactance at (60 Hz, Trefoil)	Impedance at (70 °C, 60 Hz)	Capacitance
	DC at 20 °C	AC at 70 °C			
mm²	Ω / km	Ω / km	Ω / km	Ω / km	µF / km
Non-armoured Cables					
10	3.0800	3.7007	0.139	3.703	1.028
16	1.9100	2.2950	0.130	2.299	1.254
25	1.2000	1.4420	0.125	1.447	1.271
35	0.8680	1.0432	0.119	1.050	1.489
50	0.6410	0.7706	0.113	0.779	1.497
70	0.4430	0.5329	0.108	0.544	1.753
95	0.3200	0.3854	0.106	0.400	1.782
120	0.2530	0.3052	0.103	0.322	1.964
150	0.2060	0.2490	0.102	0.269	1.955
185	0.1640	0.1989	0.101	0.223	1.969
240	0.1250	0.1527	0.100	0.182	2.042
300	0.1000	0.1233	0.098	0.158	2.113
400	0.0778	0.0975	0.097	0.137	2.189
500	0.0605	0.0779	0.095	0.123	2.318
630	0.0469	0.0630	0.093	0.113	2.580
800	0.0367	0.0524	0.091	0.105	2.907
1000	0.0291	0.0450	0.089	0.100	3.314
Armoured Cables (1)					
10	3.0800	3.7007	0.178	3.705	1.028
16	1.9100	2.2949	0.165	2.301	1.254
25	1.2000	1.4419	0.156	1.450	1.271
35	0.8680	1.0431	0.149	1.054	1.489
50	0.6410	0.7704	0.140	0.783	1.497
70	0.4430	0.5327	0.133	0.549	1.753
95	0.3200	0.3851	0.128	0.406	1.782
120	0.2530	0.3048	0.125	0.329	1.964
150	0.2060	0.2486	0.121	0.276	1.955
185	0.1640	0.1984	0.118	0.231	1.969
240	0.1250	0.1521	0.115	0.191	2.042
300	0.1000	0.1225	0.114	0.167	2.113
400	0.0778	0.0966	0.112	0.148	2.189
500	0.0605	0.0768	0.109	0.133	2.318
630	0.0469	0.0617	0.106	0.123	2.580
800	0.0367	0.0506	0.105	0.117	2.907
1000	0.0291	0.0431	0.101	0.110	3.314

(1) The tabulated values are for armoured cables with round aluminium wires (AWA) only.

Annex E: Electrical Parameters of Cables

Table E.3

Single-core cables with copper conductors – XLPE Insulated

Nominal area of conductor	Max. Conductor Resistance		Reactance at (60 Hz, Trefoil)	Impedance at (90 °C, 60 Hz)	Capacitance
	DC at 20 °C	AC at 90 °C			
mm²	Ω / km	Ω / km	Ω / km	Ω / km	μF / km
Non-armoured Cables					
1.5	12.1000	15.4287	0.175	15.430	0.212
2.5	7.4100	9.4485	0.161	9.450	0.264
4	4.6100	5.8782	0.150	5.880	0.309
6	3.0800	3.9274	0.141	3.930	0.379
10	1.8300	2.3335	0.134	2.337	0.433
16	1.1500	1.4665	0.125	1.472	0.533
25	0.7270	0.9273	0.121	0.935	0.505
35	0.5240	0.6686	0.115	0.678	0.599
50	0.3870	0.4941	0.108	0.506	0.629
70	0.2680	0.3428	0.104	0.358	0.679
95	0.1930	0.2476	0.102	0.268	0.781
120	0.1530	0.1970	0.100	0.185	0.797
150	0.1240	0.1605	0.099	0.189	0.767
185	0.0991	0.1294	0.098	0.162	0.753
240	0.0754	0.1002	0.096	0.139	0.807
300	0.0601	0.0817	0.094	0.125	0.858
400	0.0470	0.0663	0.094	0.115	0.870
500	0.0366	0.0545	0.092	0.107	0.904
630	0.0283	0.0454	0.092	0.102	0.928
800	0.0221	0.0390	0.091	0.099	0.971
1000	0.0176	0.0346	0.088	0.095	1.105
Armoured Cables (1)					
6	3.0800	3.9273	0.184	3.932	0.379
10	1.8300	2.3335	0.175	2.340	0.433
16	1.1500	1.4665	0.162	1.475	0.533
25	0.7270	0.9272	0.153	0.940	0.505
35	0.5240	0.6685	0.145	0.684	0.599
50	0.3870	0.4939	0.137	0.512	0.629
70	0.2680	0.3424	0.130	0.366	0.679
95	0.1930	0.2471	0.125	0.277	0.781
120	0.1530	0.1964	0.122	0.231	0.797
150	0.1240	0.1598	0.118	0.199	0.767
185	0.0991	0.1286	0.116	0.173	0.753
240	0.0754	0.0991	0.113	0.150	0.807
300	0.0601	0.0805	0.109	0.135	0.858
400	0.0470	0.0647	0.110	0.127	0.870
500	0.0366	0.0527	0.107	0.119	0.904
630	0.0283	0.0436	0.104	0.113	0.928
800	0.0221	0.0367	0.105	0.111	0.971
1000	0.0176	0.0324	0.100	0.105	1.105

(1) The tabulated values are for armoured cables with round aluminium wires (AWA) only.

Annex E: Electrical Parameters of Cables

Table E.4

Single-core cables with aluminium conductors - XLPE Insulated

Nominal area of conductor	Max. Conductor Resistance		Reactance at (60 Hz, Trefoil)	Impedance at (90 °C, 60 Hz)	Capacitance
	DC at 20 °C	AC at 90 °C			
mm²	Ω / km	Ω / km	Ω / km	Ω / km	μF / km
Non-armoured Cables					
10	3.0800	3.9489	0.134	3.951	0.433
16	1.9100	2.4489	0.125	2.452	0.533
25	1.2000	1.5387	0.121	1.543	0.505
35	0.8680	1.1131	0.115	1.119	0.599
50	0.6410	0.8222	0.108	0.829	0.629
70	0.4430	0.5686	0.104	0.578	0.679
95	0.3200	0.4112	0.102	0.424	0.781
120	0.2530	0.3255	0.100	0.341	0.797
150	0.2060	0.2656	0.099	0.283	0.767
185	0.1640	0.2121	0.098	0.234	0.753
240	0.1250	0.1627	0.096	0.189	0.807
300	0.1000	0.1314	0.094	0.162	0.858
400	0.0778	0.1038	0.094	0.140	0.870
500	0.0605	0.0828	0.092	0.124	0.904
630	0.0469	0.0666	0.092	0.114	0.928
800	0.0367	0.0551	0.091	0.106	0.973
1000	0.0291	0.0471	0.088	0.100	1.105
Armoured Cables (1)					
10	3.0800	3.9489	0.175	3.953	0.433
16	1.9100	2.4489	0.162	2.454	0.533
25	1.2000	1.5386	0.153	1.546	0.505
35	0.8680	1.1130	0.145	1.122	0.599
50	0.6410	0.8221	0.137	0.833	0.629
70	0.4430	0.5684	0.130	0.583	0.679
95	0.3200	0.4109	0.125	0.429	0.781
120	0.2530	0.3252	0.122	0.347	0.797
150	0.2060	0.2651	0.118	0.290	0.767
185	0.1640	0.2116	0.116	0.241	0.753
240	0.1250	0.1621	0.113	0.197	0.807
300	0.1000	0.1306	0.109	0.170	0.858
400	0.0778	0.1028	0.110	0.150	0.870
500	0.0605	0.0816	0.107	0.134	0.904
630	0.0469	0.0653	0.104	0.123	0.928
800	0.0367	0.0533	0.104	0.117	0.973
1000	0.0291	0.0452	0.100	0.110	1.105

(1) The tabulated values are for armoured cables with round aluminium wires (AWA) only.

Annex E: Electrical Parameters of Cables

Table E.5

Multi-core cables with copper conductors - PVC Insulated

Nominal area of conductor	Max. Conductor Resistance		Reactance at (60 Hz)	Impedance at (70 °C, 60 Hz)	Capacitance
	DC at 20 °C	AC at 70 °C			
mm²	Ω / km	Ω / km	Ω / km	Ω / km	µF / km
1.5	12.1000	14.4777	0.132	14.478	0.630
2.5	7.4100	8.8661	0.123	8.867	0.759
4	4.6100	5.5159	0.123	5.517	0.768
6	3.0800	3.6853	0.116	3.687	0.897
10	1.8300	2.1898	0.111	2.193	1.028
16	1.1500	1.3763	0.105	1.380	1.254
25	0.7270	0.8703	0.104	0.877	1.293
35	0.5240	0.6276	0.101	0.636	1.489
50	0.3870	0.4639	0.097	0.474	1.497
70	0.2680	0.3220	0.094	0.336	1.753
95	0.1930	0.2328	0.094	0.251	1.782
120	0.1530	0.1856	0.091	0.207	1.964
150	0.1240	0.1514	0.091	0.177	1.955
185	0.0991	0.1224	0.091	0.152	1.969
240	0.0754	0.0952	0.090	0.131	2.042
300	0.0601	0.0780	0.089	0.119	2.113
400	0.0470	0.0638	0.089	0.109	2.189
500	0.0366	0.0531	0.088	0.103	2.318

Table E.6

Multi-core cables with aluminium conductors - PVC Insulated

Nominal area of conductor	Max. Conductor Resistance		Reactance at (60 Hz)	Impedance at (70 °C, 60 Hz)	Capacitance
	DC at 20 °C	AC at 70 °C			
mm²	Ω / km	Ω / km	Ω / km	Ω / km	µF / km
10	3.0800	3.7007	0.111	3.702	1.028
16	1.9100	2.2950	0.105	2.297	1.254
25	1.2000	1.4421	0.106	1.446	1.242
35	0.8680	1.0433	0.101	1.048	1.489
50	0.6410	0.7707	0.097	0.777	1.497
70	0.4430	0.5331	0.094	0.541	1.753
95	0.3200	0.3856	0.094	0.397	1.782
120	0.2530	0.3055	0.091	0.319	1.964
150	0.2060	0.2494	0.091	0.265	1.955
185	0.1640	0.1994	0.091	0.219	1.969
240	0.1250	0.1533	0.090	0.178	2.042
300	0.1000	0.1240	0.090	0.153	2.113
400	0.0778	0.0984	0.089	0.132	2.189
500	0.0605	0.0789	0.088	0.118	2.318

Annex E: Electrical Parameters of Cables

Table E.7

Multi-core cables with copper conductors - XLPE Insulated

Nominal area of conductor	Max. Conductor Resistance		Reactance at (60 Hz)	Impedance at (90 °C, 60 Hz)	Capacitance
	DC at 20 °C	AC at 90 °C			
mm²	Ω / km	Ω / km	Ω / km	Ω / km	µF / km
1.5	12.1000	15.4287	0.128	15.429	0.212
2.5	7.4100	9.4485	0.118	9.449	0.264
4	4.6100	5.8783	0.113	5.879	0.309
6	3.0800	3.9274	0.106	3.929	0.379
10	1.8300	2.3336	0.103	2.336	0.433
16	1.1500	1.4667	0.098	1.470	0.533
25	0.7270	0.9275	0.099	0.933	0.505
35	0.5240	0.6688	0.096	0.676	0.599
50	0.3870	0.4944	0.092	0.503	0.629
70	0.2680	0.3431	0.090	0.355	0.679
95	0.1930	0.2481	0.088	0.263	0.781
120	0.1530	0.1976	0.087	0.216	0.797
150	0.1240	0.1612	0.088	0.183	0.767
185	0.0991	0.1302	0.088	0.157	0.753
240	0.0754	0.1012	0.087	0.133	0.807
300	0.0601	0.0829	0.086	0.119	0.858
400	0.0470	0.0676	0.085	0.109	0.870
500	0.0366	0.0561	0.085	0.102	0.904

Table E.8

Multi-core cables with aluminium conductors - XLPE Insulated

Nominal area of conductor	Max. Conductor Resistance		Reactance at (60 Hz)	Impedance at (90 °C, 60 Hz)	Capacitance
	DC at 20 °C	AC at 90 °C			
mm²	Ω / km	Ω / km	Ω / km	Ω / km	µF / km
10	3.0800	3.9490	0.103	3.950	0.433
16	1.9100	2.4490	0.098	2.451	0.533
25	1.2000	1.5388	0.099	1.542	0.497
35	0.8680	1.1133	0.096	1.117	0.599
50	0.6410	0.8224	0.092	0.827	0.629
70	0.4430	0.5688	0.090	0.576	0.679
95	0.3200	0.4115	0.088	0.421	0.781
120	0.2530	0.3259	0.087	0.337	0.797
150	0.2060	0.2660	0.088	0.280	0.767
185	0.1640	0.2126	0.088	0.230	0.753
240	0.1250	0.1634	0.087	0.185	0.807
300	0.1000	0.1321	0.086	0.158	0.858
400	0.0778	0.1047	0.085	0.135	0.870
500	0.0605	0.0838	0.085	0.119	0.904

Annex F: Short Circuit Capacity

F.1 Permissible short circuit current

The short circuit capacity of a current carrying component of a cable is determined by the following factors:

- The temperature prior to the short circuit, generally taken to be that corresponding with the maximum conductor operating temperature under normal conditions
- The energy produced by the short circuit, a function of both the magnitude and the duration of the current
- The limiting final temperature, generally determined by all materials in direct contact with the conducting component

The short circuit current ratings given below in Tables F.1 to F.4 are calculated in accordance with the following formula as given in IEC 60949:

$$I = \frac{K\ x\ S}{\sqrt{t}} \sqrt{Ln\left(\frac{\theta_f\ +\ \beta}{\theta_i\ +\ \beta}\right)}$$

Where

I : Permissible Short circuit current *(A)*

t : Duration of short circuit *(seconds)*

S : Cross-sectional area of the current-carrying component *(mm^2)*

K : Constant depending on the material of the current-carrying component *($As^{½} / mm^2$)*

θ_i : Initial temperature before short circuit in *(°C)*

θ_f : Final temperature at short circuit in *(°C)*

β : Reciprocal of temperature coefficient of resistance of the current carrying component at 0 °C

The short-circuit current ratings given below are the symmetrical currents which will cause the conductor temperature to rise from the normal operating value to the maximum short-circuit value in the time stated, assuming adiabatic conditions (i.e. neglecting heat loss).

Annex F: Short Circuit Capacity

Table F.1

Short circuit current (kA) – Copper conductor – PVC Insulated

Nominal area of conductor	Short circuit duration (sec.)									
mm^2	0.1	0.2	0.3	0.4	0.5	1.0	2.0	3.0	4.0	5.0
1.5	0.55	0.39	0.31	0.27	0.24	0.17	0.12	0.10	0.09	0.08
2.5	0.91	0.64	0.52	0.45	0.41	0.29	0.20	0.17	0.14	0.13
4	1.45	1.03	0.84	0.73	0.65	0.46	0.33	0.27	0.23	0.21
6	2.18	1.54	1.26	1.09	0.98	0.69	0.49	0.40	0.35	0.31
10	3.6	2.6	2.1	1.8	1.6	1.2	0.8	0.7	0.6	0.5
16	5.8	4.1	3.4	2.9	2.6	1.8	1.3	1.1	0.9	0.8
25	9.1	6.4	5.2	4.5	4.1	2.9	2.0	1.7	1.4	1.3
35	12.7	9.0	7.3	6.4	5.7	4.0	2.8	2.3	2.0	1.8
50	18.2	12.9	10.5	9.1	8.1	5.8	4.1	3.3	2.9	2.6
70	25.5	18.0	14.7	12.7	11.4	8.1	5.7	4.6	4.0	3.6
95	34.5	24.4	19.9	17.3	15.5	10.9	7.7	6.3	5.5	4.9
120	43.6	30.9	25.2	21.8	19.5	13.8	9.8	8.0	6.9	6.2
150	54.5	38.6	31.5	27.3	24.4	17.3	12.2	10.0	8.6	7.7
185	67.3	47.6	38.8	33.6	30.1	21.3	15.0	12.3	10.6	9.5
240	87.3	61.7	50.4	43.6	39.0	27.6	19.5	15.9	13.8	12.3
300	109.1	77.1	63.0	54.5	48.8	34.5	24.4	19.9	17.3	15.4
400	130.0	91.9	75.1	65.0	58.2	41.1	29.1	23.7	20.6	18.4
500	162.5	114.9	93.8	81.3	72.7	51.4	36.3	29.7	25.7	23.0
630	204.8	144.8	118.2	102.4	91.6	64.8	45.8	37.4	32.4	29.0
800	260.1	183.9	150.1	130.0	116.3	82.2	58.2	47.5	41.1	36.8
1000	325.1	229.9	187.7	162.5	145.4	102.8	72.7	59.4	51.4	46.0

Table F.2

Short circuit current (kA) – Aluminum conductor – PVC Insulated

Nominal area of conductor	Short circuit duration (sec.)									
mm^2	0.1	0.2	0.3	0.4	0.5	1.0	2.0	3.0	4.0	5.0
10	2.40	1.70	1.39	1.20	1.07	0.76	0.54	0.44	0.38	0.34
16	3.8	2.7	2.2	1.9	1.7	1.2	0.9	0.7	0.6	0.5
25	6.0	4.2	3.5	3.0	2.7	1.9	1.3	1.1	1.0	0.8
35	8.4	5.9	4.9	4.2	3.8	2.7	1.9	1.5	1.3	1.2
50	12.0	8.5	6.9	6.0	5.4	3.8	2.7	2.2	1.9	1.7
70	16.8	11.9	9.7	8.4	7.5	5.3	3.8	3.1	2.7	2.4
95	22.8	16.1	13.2	11.4	10.2	7.2	5.1	4.2	3.6	3.2
120	28.8	20.4	16.7	14.4	12.9	9.1	6.4	5.3	4.6	4.1
150	36.0	25.5	20.8	18.0	16.1	11.4	8.1	6.6	5.7	5.1
185	44.5	31.4	25.7	22.2	19.9	14.1	9.9	8.1	7.0	6.3
240	57.7	40.8	33.3	28.8	25.8	18.2	12.9	10.5	9.1	8.2
300	72.1	51.0	41.6	36.0	32.2	22.8	16.1	13.2	11.4	10.2
400	86.0	60.8	49.7	43.0	38.5	27.2	19.2	15.7	13.6	12.2
500	107.5	76.0	62.1	53.8	48.1	34.0	24.0	19.6	17.0	15.2
630	135.5	95.8	78.2	67.7	60.6	42.8	30.3	24.7	21.4	19.2
800	172.0	121.6	99.3	86.0	76.9	54.4	38.5	31.4	27.2	24.3
1000	215.0	152.1	124.2	107.5	96.2	68.0	48.1	39.3	34.0	30.4

Annex F: Short Circuit Capacity

Table F.3

Short circuit current (kA) – Copper conductor – XLPE Insulated

Nominal area of conductor	Short circuit duration (sec.)									
mm^2	0.1	0.2	0.3	0.4	0.5	1.0	2.0	3.0	4.0	5.0
1.5	0.68	0.48	0.39	0.34	0.30	0.21	0.15	0.12	0.11	0.10
2.5	1.13	0.80	0.65	0.57	0.51	0.36	0.25	0.21	0.18	0.16
4	1.81	1.28	1.04	0.90	0.81	0.57	0.40	0.33	0.29	0.26
6	2.71	1.92	1.57	1.36	1.21	0.86	0.61	0.50	0.43	0.38
10	4.52	3.20	2.61	2.26	2.02	1.43	1.01	0.83	0.72	0.64
16	7.2	5.1	4.2	3.6	3.2	2.3	1.6	1.3	1.1	1.0
25	11.3	8.0	6.5	5.7	5.1	3.6	2.5	2.1	1.8	1.6
35	15.8	11.2	9.1	7.9	7.1	5.0	3.5	2.9	2.5	2.2
50	22.6	16.0	13.1	11.3	10.1	7.2	5.1	4.1	3.6	3.2
70	31.7	22.4	18.3	15.8	14.2	10.0	7.1	5.8	5.0	4.5
95	43.0	30.4	24.8	21.5	19.2	13.6	9.6	7.8	6.8	6.1
120	54.3	38.4	31.3	27.1	24.3	17.2	12.1	9.9	8.6	7.7
150	67.9	48.0	39.2	33.9	30.4	21.5	15.2	12.4	10.7	9.6
185	83.7	59.2	48.3	41.9	37.4	26.5	18.7	15.3	13.2	11.8
240	108.6	76.8	62.7	54.3	48.6	34.3	24.3	19.8	17.2	15.4
300	135.7	96.0	78.4	67.9	60.7	42.9	30.4	24.8	21.5	19.2
400	181.0	128.0	104.5	90.5	80.9	57.2	40.5	33.0	28.6	25.6
500	226.2	160.0	130.6	113.1	101.2	71.5	50.6	41.3	35.8	32.0
630	285.1	201.6	164.6	142.5	127.5	90.1	63.7	52.0	45.1	40.3
800	362.0	256.0	209.0	181.0	161.9	114.5	80.9	66.1	57.2	51.2
1000	452.5	319.9	261.2	226.2	202.4	143.1	101.2	82.6	71.5	64.0

Table F.4

Short circuit current (kA) – Aluminum conductor – XLPE Insulated

Nominal area of conductor	Short circuit duration (sec.)									
mm^2	0.1	0.2	0.3	0.4	0.5	1.0	2.0	3.0	4.0	5.0
10	2.99	2.11	1.73	1.49	1.34	0.94	0.67	0.55	0.47	0.42
16	4.8	3.4	2.8	2.4	2.1	1.5	1.1	0.9	0.8	0.7
25	7.5	5.3	4.3	3.7	3.3	2.4	1.7	1.4	1.2	1.1
35	10.5	7.4	6.0	5.2	4.7	3.3	2.3	1.9	1.7	1.5
50	14.9	10.6	8.6	7.5	6.7	4.7	3.3	2.7	2.4	2.1
70	20.9	14.8	12.1	10.5	9.4	6.6	4.7	3.8	3.3	3.0
95	28.4	20.1	16.4	14.2	12.7	9.0	6.3	5.2	4.5	4.0
120	35.9	25.4	20.7	17.9	16.0	11.3	8.0	6.5	5.7	5.1
150	44.8	31.7	25.9	22.4	20.0	14.2	10.0	8.2	7.1	6.3
185	55.3	39.1	31.9	27.6	24.7	17.5	12.4	10.1	8.7	7.8
240	71.7	50.7	41.4	35.9	32.1	22.7	16.0	13.1	11.3	10.1
300	89.6	63.4	51.8	44.8	40.1	28.3	20.0	16.4	14.2	12.7
400	119.5	84.5	69.0	59.8	53.4	37.8	26.7	21.8	18.9	16.9
500	149.4	105.6	86.3	74.7	66.8	47.2	33.4	27.3	23.6	21.1
630	188.2	133.1	108.7	94.1	84.2	59.5	42.1	34.4	29.8	26.6
800	239.0	169.0	138.0	119.5	106.9	75.6	53.4	43.6	37.8	33.8
1000	298.8	211.3	172.5	149.4	133.6	94.5	66.8	54.6	47.2	42.3

Annex G : Voltage Drop Calculations

G.1 Voltage drop

When current flows in a cable conductor, there is a voltage drop between the ends of the conductor, which is the product of the current and the impedance. If the voltage drop were excessive, it could result in the voltage at the equipment being supplied being too low for proper operation.

In order to ensure satisfactory operation of electrical equipment, it is necessary to maintain the voltage at which it is supplied within certain limits.

G.2 Voltage drop limitation in consumers' installations

Most of Electrical Codes normally permit a 3% voltage drop in a "feeder" or "branch circuit" and no more than 5% voltage drop from the supply side of the consumer's service to the point of utilization.

The voltage drop limitation applying to a circuit needs to be assessed taking account of the function of the circuit and its relationship with other circuits. For example, other voltage drop limits may apply in ELV circuits or may be dictated by motor starting considerations.

In addition, the voltage drop in mains and sub-mains circuits should take account of the voltage drop in final sub-circuits (and vice versa) to ensure the total voltage drop in the installation is within the required limits.

G.3 Use of tabulated mV/A/m values

The process of calculating the voltage drop of a cable under a given operating conditions is normally straightforward. The tabulated voltage drop (mV/A/m) values in Tables G.1 & G.2 represent the worst case conditions, whereby it is assumed that the cable is operating at its maximum permitted operating temperature (70 and 90 °C for PVC and XLPE insulated cables respectively) and is supplying a load having a power factor equal to the power factor of the cable. For three-phase circuits, balanced loading is assumed.

On this basis, where the cable size and type, load current and length of run are known, the voltage drop can be calculated from the below formula. This formula is used to calculate the voltage drop in a circuit when the cable size is known.

$$V_d = \frac{V_t \, x \, I \, x \, L}{1000} \; (V)$$

Where

V_d : The voltage drop for the cable run (V)
V_t : The tabulated voltage drop value for the cable (mV/A/m)
I : The load current (A)
L : The length of cable route/run (m)

G.4 Example

Assume a 4-core 10 mm^2 cable with copper conductors and XLPE insulated, carrying a load current of 20 A over 60 m route length.

At normal operation, the voltage drop would be:

3.53 x 20 x 60 x 0.001 = 4.236 V

Where 3.53 is the tabulated value in Table G.2 for the voltage drop per amp per meter (mV/A/m) of this cable at its maximum permitted operating temperature (90 °C).

Annex G: Voltage Drop Values

Table G.1

Voltage drop for PVC insulated cables

Nominal area of conductor	Voltage drop (mV/Amp/Meter)									
	Single-core cables								Multi-core cables	
	Unarmoured cables				Armoured cables					
mm²	Copper		Aluminum		Copper		Aluminum		Copper	Aluminum
	Flat	Trefoil	Flat	Trefoil	Flat	Trefoil	Flat	Trefoil		
1.5	21.494	21.478	-	-	21.540	21.524	-	-	21.436	-
2.5	13.219	13.203	-	-	13.263	13.247	-	-	13.166	-
4	8.281	8.265	-	-	8.320	8.304	-	-	8.233	-
6	5.576	5.560	-	-	5.614	5.598	-	-	5.532	-
10	3.367	3.351	5.592	5.576	3.403	3.387	5.627	5.611	3.326	5.550
16	2.161	2.145	3.514	3.498	2.193	2.177	3.546	3.530	2.123	3.475
25	1.412	1.396	2.253	2.237	1.440	1.424	2.282	2.266	1.377	2.220
35	1.049	1.033	1.661	1.645	1.076	1.060	1.688	1.672	1.017	1.629
50	0.802	0.786	1.254	1.238	0.827	0.811	1.279	1.263	0.772	1.224
70	0.588	0.572	0.899	0.883	0.611	0.595	0.922	0.906	0.561	0.871
95	0.456	0.440	0.681	0.665	0.475	0.459	0.700	0.684	0.429	0.654
120	0.383	0.367	0.560	0.544	0.402	0.386	0.579	0.563	0.357	0.533
150	0.331	0.315	0.476	0.460	0.348	0.332	0.493	0.477	0.307	0.451
185	0.288	0.272	0.402	0.386	0.302	0.286	0.417	0.401	0.264	0.377
240	0.246	0.230	0.332	0.316	0.259	0.243	0.346	0.330	0.223	0.309
300	0.219	0.203	0.287	0.271	0.232	0.216	0.301	0.285	0.197	0.265
400	0.197	0.181	0.249	0.233	0.209	0.193	0.261	0.245	0.175	0.226
500	0.179	0.163	0.218	0.202	0.189	0.173	0.229	0.213	0.159	0.197
630	0.165	0.149	0.194	0.179	0.174	0.158	0.204	0.188	-	-
800	0.155	0.139	0.177	0.161	0.164	0.148	0.187	0.171	-	-
1000	0.147	0.131	0.164	0.148	0.154	0.138	0.172	0.156	-	-

NOTE: *The above values for cables voltage drop per ampere per meter were calculated based on the following conditions*

- Maximum permitted operating temperature for conductor : 70 °C
- Power factor : 0.85
- Rated frequency : 60 Hz

Annex G: Voltage Drop Values

Table G.2

Voltage drop for XLPE insulated cables

Nominal area of conductor	Voltage drop (mV/Amp/Meter)									
	Single-core cables								Multi-core cables	
	Unarmoured cables				Armoured cables					
mm²	Copper		Aluminum		Copper		Aluminum		Copper	Aluminum
	Flat	Trefoil	Flat	Trefoil	Flat	Trefoil	Flat	Trefoil		
1.5	22.892	22.876	-	-	22.939	22.923	-	-	22.832	-
2.5	14.074	14.058	-	-	14.120	14.104	-	-	14.019	-
4	8.808	8.792	-	-	8.850	8.834	-	-	8.757	-
6	5.927	5.911	-	-	5.967	5.951	-	-	5.880	-
10	3.574	3.558	5.952	5.936	3.612	3.596	5.990	5.974	3.530	5.908
16	2.289	2.273	3.736	3.720	2.324	2.308	3.770	3.754	2.249	3.696
25	1.492	1.476	2.392	2.376	1.522	1.506	2.422	2.406	1.456	2.357
35	1.106	1.090	1.760	1.744	1.134	1.118	1.788	1.772	1.073	1.727
50	0.843	0.827	1.326	1.310	0.869	0.853	1.352	1.336	0.812	1.295
70	0.617	0.601	0.949	0.933	0.639	0.623	0.972	0.956	0.588	0.920
95	0.474	0.458	0.715	0.699	0.494	0.478	0.735	0.719	0.446	0.687
120	0.398	0.382	0.587	0.571	0.417	0.401	0.607	0.591	0.371	0.560
150	0.343	0.327	0.498	0.482	0.360	0.344	0.515	0.499	0.318	0.472
185	0.296	0.280	0.418	0.402	0.312	0.296	0.434	0.418	0.272	0.394
240	0.252	0.236	0.344	0.328	0.265	0.249	0.358	0.342	0.229	0.320
300	0.223	0.207	0.296	0.280	0.235	0.219	0.308	0.292	0.201	0.273
400	0.199	0.183	0.255	0.239	0.212	0.196	0.268	0.252	0.178	0.232
500	0.181	0.165	0.223	0.207	0.192	0.176	0.234	0.218	0.160	0.201
630	0.167	0.151	0.198	0.182	0.176	0.160	0.208	0.192	-	-
800	0.157	0.141	0.180	0.164	0.166	0.150	0.190	0.174	-	-
1000	0.148	0.132	0.166	0.150	0.156	0.140	0.175	0.159	-	-

NOTE: *The above values for cables voltage drop per ampere per meter were calculated based on the following conditions*

- Maximum permitted operating temperature for conductor : 90 °C
- Power factor : 0.85
- Rated frequency : 60 Hz

Annex H : Cable Sizing Calculations

H.1 Cable Sizing Calculations

As highlighted earlier, it is of vital importance that the type and size of cable ordered is suitable for its intended use. The proper sizing of an electric cable is important to ensure that the cable can:

- Operate continuously under full load without being damaged or degraded
- Withstand the worst short-circuit currents flowing through the cable
- Provide the load with a suitable voltage (and avoid excessive voltage drops)

Cable sizing methods do differ across international standards (e.g. IEC, NEC, BS, etc.) and some standards emphasize certain things over others. However, the general principles underlying any cable sizing calculations do not change. The main three factors that principally affect the minimum size of cable required for a particular installation are:

1. **The cable current-carrying capacity (load current)**, which is influenced by the cable materials, constrution, and installation conditions.
2. **The voltage drop in the cable circuit**, which is a function of load current, load power factor, and length of the cable run.
3. **The short-circuit current**, which is a function of both the magnitude and duration of the short-circuit current and is limited by the cable materials.

H.2 General Methodology for the Selection of Conductor Size

To select the proper cable size, based on the above three considerations, all cable sizing methods more or less follow the same basic five steps process.

1. Gathering data about the cable, its installation conditions, the load that it will carry, etc.
2. Determine the minimum cable size based on continuous current-carrying capacity
3. Determine the minimum cable size based on voltage drop considerations
4. Determine the minimum cable size based on short-circuit current
5. Select the cable based on the highest of the sizes calculated in step 2, 3, and 4

H.2.1 Step 1: Data Gathering

The first step is to collect the relevant information that is required to perform the sizing calculation. Typically, you will need to obtain the following data:

H.2.1.1 Load Details

The characteristics of the load that the cable will supply, which includes:

- Load type: motor or feeder
- Three-phase, single-phase or DC
- System / source voltage
- Full load current (A) - or calculate this if the load is defined in terms of power (kW)
- Full load power factor
- Distance / length of cable run from source to load

H.2.1.2 Cable Construction

The basic characteristics of the cable's physical construction, which includes:

- Number of cores – single-core or multi-core (e.g. 2C, 3C or 4C)
- Conductor material - e.g. copper or aluminium
- Insulation type - e.g. PVC or XLPE
- Armour type, if required - e.g. AWA, SWA or STA

H.2.1.3 Installation Conditions

Where and how the cable will be installed, which includes:

- Above ground or underground
- Installation / arrangement - e.g. for underground cables, is it directly buried or buried in conduit? For above ground cables, is it installed on cable tray / ladder, against a wall, in air, etc.?
- Ambient or soil temperature of the installation site
- Cable grouping, i.e. the number of cables that are installed together in one group
- Cable spacing, i.e. whether cables are installed touching or spaced
- Soil thermal resistivity (for underground cables)
- Depth of laying (for underground cables)
- For single-core three-phase cables, are the cables installed in trefoil or laid flat?

Annex H : Cable Sizing Calculations

H.2.2 Step 2: Selection of conductor size based on current rating (i.e. load current):

H.2.2.1 Determine the minimum current for which the cable is to be rated, i.e. the maximum load current.
H.2.2.2 Ascertain the actual installation conditions of the cable. In case the actual installation conditions are different from the standard laying conditions, appropriate correction (derating) factors from Annex C have to be applied to cater for the actual installation conditions.
H.2.2.3 Divide the rating from step (H.2.2.1) by the appropriate factor(s) from step (H.2.2.2).
H.2.2.4 From the tabulated Continuous Current Ratings of the required cable type and intended installation method, select a cable, which has a tabulated rating not less than the value obtained from step (H.2.2.3).

H.2.3 Step 3: Selection of conductor size based on voltage drop:

H.2.3.1 Determine the maximum Load Current I (A) to be carried by the cable, and the Route Length L (m) of the circuit.
H.2.3.2 Calculate the maximum voltage drop V_d (V) permitted in the circuit (say 3.0 % of 380 or 400 V).
H.2.3.3 Using the below equation, calculate the value of V_{ap} (mV/A/m) necessary to meet the specific voltage drop value V_d, calculated in step (H.2.3.2).

$$V_{ap} = \frac{Vd \ x \ 1000}{I \ x \ L} \quad (mV/A/m)$$

Where

V_{ap} : The maximum voltage drop value permitted for the cable (mV/A/m)
V_d : The maximum voltage drop value permitted in the circuit (V)
I : The load current (A)
L : The length of cable route/run (m)

H.2.3.4 From the tabulated voltage drop (mV/A/m) values of the required cable type in Tables G.1 & G.2, select a cable, which has a tabulated voltage drop (mV/A/m) value equal or less than the value obtained from step (H.2.3.3).

H.2.4 Step 4: Selection of conductor size based on short-circuit current:

H.2.4.1 Determine the duration t (s) and the magnitude I_{sc} (A) of the prospective short-circuit current.
H.2.4.2 Based on the required cable type and duration t (s) of short-circuit, select a cable from the tabulated short circuit current values in Tables F.1, F.2, F.3 & F.4, which has a tabulated short-circuit current value not less than the value determined in step (H.2.4.1).

H.2.5 Step 5: Determination of conductor size:

For any circuit, the cable size selected should not be less than the largest of the sizes calculated to meet the above limitations (this is the smallest size which will meet all of the requirements).

In practice, the current-carrying capacity will be found to prevail in short-run/high-current circuits while voltage drop considerations will usually prevail in long-run/low-current circuits. It is unusual for short-circuit current requirements to determine the conductor size required for low voltage cable

Annex H : Cable Sizing Calculations

H.3 Calculation Example

In this example, we will size a cable for a **40kW, 400V** three-phase motor from the Motor Control Center (MCC) to the field.

H.3.1 Step 1: Data Gathering

The first step is to collect the relevant information that is required to perform the sizing calculation. The following data was collected for the cable to be sized:

- Cable type: Cu/XLPE/SWA/PVC, 3C+E, 0.6/1kV
- Cable installation: above ground on cable ladder bunched (touched) together with 3 other cables on a single layer and at **30 °C** ambient temperature
- Cable run: **80m** (including tails)
- Motor load: **40kW, 400V** three-phase, full load current = **68A**, power factor = **0.85**
- The maximum permissible voltage drop in the circuit = **3.0 %**.
- Protection: maximum prospective short-circuit current (I_{sc}) = **0.6 kA**, duration (t) = **3.0 sec.**

H.3.2 Step 2: Selection of conductor size based on current rating (i.e. load current):

H.3.2.1 The motor full load current = 68A.

H.3.2.2 From Annex (C): The ambient temperature derating is 1.10 (Table C.2), and the grouping derating for 4 bunched (touched) cables on a cable ladder is 0.80 (Table C.8). The overall derating factor is 1.10 X 0.80 = 0.88.

H.3.2.3 The motor full load current after derating is 68/0.88 = 78A, i.e. the current rating for the selected cable should not be less than 78A.

H.3.2.4 From the tabulated current ratings of four-core cables laid in free air for (Cu/XLPE/SWA/PVC) cable type, 4 X 16 mm^2 cable has a tabulated current rating value of 96A, which exceeds the motor full load current (78A). **Therefore, 4 X 16 mm^2 is the minimum cable size that should be selected based on load current consideration.**

H.3.3 Step 3: Selection of conductor size based on voltage drop:

H.3.3.1 The maximum load current (I) = 78A, and the cable route length (L) = 80m.

H.3.3.2 The maximum permissible voltage drop in the circuit (V_d) = 3.0 % = 12V.

H.3.3.3 Using the equation stated in (H.2.3.3), V_{ap} = 1.923 mV/A/m.

H.3.3.4 From the tabulated voltage drop (mV/A/m) values of the required cable type (Table G.2), the voltage drop value of 16 mm^2 cable (2.249 mV/A/m) is higher than the value calculated for V_{ap}, and therefore is not complying with the maximum permissible voltage drop requirement. The 25 mm^2 cable on the other hand has a tabulated voltage drop value of 1.456 mV/A/m, which is lower than the value calculated for V_{ap}. **Therefore, 4 X 25 mm^2 is the minimum cable size that should be selected based on maximum permissible voltage drop consideration.**

H.3.4 Step 4: Selection of conductor size based on short-circuit current:

H.3.4.1 The magnitude of the prospective short-circuit current (I_{sc}) = 0.6kA, and the duration (t) = 3.0 Sec.

H.3.4.2 From the tabulated short-circuit current values for Cu/XLPE cable type (Table F.3), 10 mm^2 cable has a tabulated short-circuit current value of 0.83kA, which exceeds the maximum prospective short-circuit current of the system (0.6kA). **Therefore, 4 X 10 mm^2 is the minimum cable size that should be selected based on short-circuit current consideration.**

H.3.5 Step 5: Determination of conductor size:

For any circuit, the cable size selected should not be less than the largest of the sizes calculated to meet the above limitations (this is the smallest size which will meet all of the requirements).

Therefore, 4 X 25 mm^2 is the minimum cable size that should be selected to fulfill all the above requirements.

Annex I: Cable Installation Practice

I.1 General

This annex provides installation methods commonly encountered in industrial, residential and electric power utility applications and should be used in conjunction with the engineer's installation specifications and all applicable codes/standards.

The cables included in this catalogue are designed to be installed in air, or for burial in free draining soil conditions either direct buried in the ground or in buried ducts. Where the cables are to be laid in any other environment, reference should be made to **alfanar** for proper recommendations.

I.1.1 Single-core Cables

The following points relating to single-core cables should be noted:

1. Single-core cables carrying the phase currents of a three-phase a.c. circuit must be installed as closely as possible together, to minimize inductive reactance and voltage drop. The preferred formation for three-phase conductors is a "trefoil" or cloverleaf pattern although flat touching formation is also acceptable. Sheaths should be in contact with one another in either case.
2. A single-core cable generates an alternating magnetic field around itself which can cause large increases in voltage drop and power loss due to "transformer effect" when ferrous metal (iron and steel) is allowed to encircle the cable. Steel racking or ladder will not induce this effect, but the following must be observed:
    a. Cable cleats may be of wood, plastic, or non-ferrous metal but steel saddles should not be used on single cores.
    b. Where three single phase cables pass through a steel bulkhead, they must all pass through the same hole Where glanding is required, it is usual to cut out a panel and replace this with a non-ferrous (metal or plastic) plate in which the three or four glands are mounted.

I.1.2 Lugs and Links

Stranded compacted conductors, either round or sector shaped, must have lugs and links fitted that are manufactured for the same nominal cross-sectional area as the conductor. For example, a 150 mm^2 conductor must have a 150 mm^2 lug or link fitted, and the correct dies, as stated by the manufacturer, should be used to compress it.

Although the lug or link will appear to be loose on the conductor, this is simply because the initial compression of the joint has already taken place during the manufacture of the conductor; the final compression of the joint will be correct.

If, for example, a 120 mm^2 lug or link was fitted to a 150 mm^2 conductor, the joint would be over compressed and likely to fail in service. In addition, the smaller lug in itself would be unable to carry the same maximum current as the larger conductor, particularly with respect to fault currents.

I.2 Pre-installation

To ensure safety during cable installation and reliability once the cable is installed, you should confirm the following prior to installation.

- The cable selected is proper for your application
- The cable has not been damaged in transit or storage

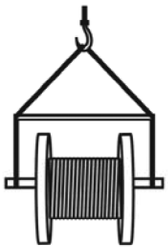
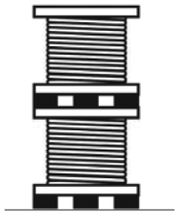
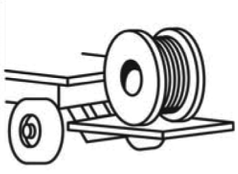
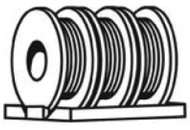
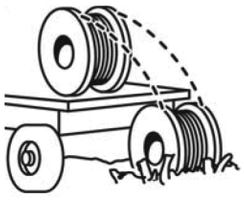
Review all applicable local, state, provincial, and national codes to verify that the cable selected is appropriate for the installation job.

Any existing cable damage must be identified and any further damage prevented from occurring. This is done through proper cable inspection, handling and storage.

I.2.1 Drum Handling

The following recommendations for the storage and handling of cables that are packed on wooden cable drums suitable for storage outdoors should be followed.

Annex I: Cable Installation Practice

Yes	No
	
Cradle both reel flanges between forks	Do not lift by top flange. Cable or reel will be damaged
	
Reels can be hoisted with a shaft extended through both flanges	Use a spreader bar to prevent bending the reel flanges and mashing the cable
	
Place spacers under the bottom flange and between reels to create a space to insert the forks	Upended heavy reels will often arrive damaged. Refuse or receive subject to inspection for hidden damage
	
Lower reels from truck using hydraulic gate, hoist or fork lift. LOWER CAREFULLY.	Never allow forks to touch cable surface or reel wrap
	
Always load with flanges on edge and chock and block securely	Never drop reels

Annex I: Cable Installation Practice

I.2.2 Cable Inspection

Inspect every reel of cable for damage before accepting the shipment. Be particularly alert for cable damage if:

- A reel is laying flat on its flange side
- Several reels are stacked on top of each other
- Other freight is stacked on top of a reel
- Nails have been driven into reel flanges to secure shipping blocks
- A reel flange is damaged
- A reel has been dropped (hidden damage likely)
- A cable covering has been removed, or is stained or damaged
- A cable end seal has been removed or is damaged

I.2.3 Cable Storage

Cables should be stored on hard surfaces so that reel flanges cannot sink. Small reels may weight several hundred kilograms while large reels can exceed several thousand kilograms.

Impact damage can be prevented by the following precautions:

- Aligning reels flange to flange
- Using guards across flanges when different reel sizes are stored together
- Maintaining adequate aisles and barricades to prevent equipment from hitting the cable

I.2.4 Prevention of moisture ingress

Care should be exercised during installation to avoid any damage to cable coverings. This is important in wet or other aggressive environments. The protective end cap should not be removed from the ends of the cable until immediately prior to termination or jointing, especially for cables that do not have an extruded bedding. When the caps have been removed, the unprotected ends of the cable should not be exposed to moisture.

I.3 Installation

A high percentage of cable failures are due to mechanical damage, which typically occurs during transportation, handling and installation.

In fact, most cables are subjected to more mechanical stress during installation than they ever experience in actual operation. Needless to say, handling and installing the cable according to the manufacturer's recommendations is extremely important.

When cables are installed in a raceway, underground electrical duct or cable tray, the following factors must be considered:

- Conductor configuration
- Raceway or cable tray fill
- Physical limitations of cables
- Installation equipment
- Ambient temperature and conditions

Similarly, when cable is installed as exposed wiring or as messenger-supported wiring, all of the above factors except raceway or cable tray fill must be considered as well as the requirements for securing and supporting the cables.

I.3.1 Minimum installation radius

None of the cables included in this catalogue should be bent during installation to a radius smaller than the following:

Single-core cables
- Unarmoured : 15 Ø
- Armoured : 12 Ø

Multi-core cables
- Unarmoured : 12 Ø
- Armoured : 12 Ø

Where Ø is the overall diameter of the cable.

Wherever possible, larger installation radius should be used, except that the minimum bending radius where the cables are placed in position adjacent to joints and terminations may be reduced to that values given below, provided that the bending is carefully controlled, e.g. by the use of a former.

Single-core cables
- Unarmoured : 8 Ø
- Armoured : 8 Ø

Multi-core cables
- Unarmoured : 6 Ø
- Armoured : 6 Ø

I.3.2 Minimum temperature during installation

It is recommended that the cables should be installed only when both the cable and ambient temperature are above 0 °C and have been so for the previous 24 hours, or where special precautions have been taken to maintain the cable above this temperature.

I.3.3 Maximum pulling tension

The maximum pulling tension is depending on the cable design, the mechanical limitations, the conductor material, and the method of laying and pulling the cables. The maximum permissible pulling force can be calculated based on the method of pulling as follows:

I.3.3.1 Pulling eye attached to the conductor

With pulling eye attached to copper conductors, the maximum pulling tension should not exceed 0.036 times circular-mil area of conductor (C_m). With pulling eye attached to aluminum conductors, the maximum pulling tension should not exceed 0.027 times circular-mil area of conductor (C_m). Or in other words

$T_m = 0.036 \ X \ n \ X \ C_m$ (Copper)

$T_m = 0.027 \ X \ n \ X \ C_m$ (Aluminum)

Where

T_m : Maximum pulling tension in N
n : The number of conductors
C_m : Circular mil area of each conductor

The maximum limitation for this calculation is **22240 N** (2268 kgf) for single conductor (1/C) cables, and **44480** N (4536 kgf) for multi core cables. This limitation is due to unequal distribution of tension forces when pulling multiple conductors.

When the calculated pulling tension is close to (or within 10 % of) the maximum pulling tension, the use of a tension gauge during the pulling is recommended.

I.3.3.2 Cable grip over lead sheath

With cable grip over lead sheath, with commercial lead, the maximum pulling tension on the lead sheath should not exceed **10.33 N/mm²** (1500 lbf/in^2).

I.3.3.3 Cable grip over non-leaded cable

With cable grip over non-leaded cable, the maximum pulling tension should not exceed 4400 N (1000 lbf).

Annex I: Cable Installation Practice

I.3.4 Side wall pressure

One of the limitations to be considered in the installation of electrical cables is sidewall pressure. The sidewall pressure is the force exerted on the insulation and sheath of the cable at a bend point when the cable is under tension, and is normally the limiting factor in an installation where cable bends are involved. The sidewall pressure in general is expressed as the tension out of a bend expressed in newtons divided by the inside radius of the bend expressed in meters.

$$P = \left[\frac{T_o}{r} \right]$$

Where

P : Sidewall pressure in N/m (lbf/ft)
T_o : Tension leaving the bend in N (lbf)
r : Inside radius of conduit in m (ft)

The normal maximum sidewall pressure per meter (foot) of radius is as given below. However, in order to minimize cable damage because of excessive sidewall pressure, the installer should check the proper recommendations for each type of cables to be installed.

Cable Type	Maximum Sidewall Pressure	
	(N/m)	(lbf/ft)
Non-shielded multi-core cables	7300	500
Single-core cables	7300	500
Armoured cables	4400	300

Annex J: Coding Key

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18

The type designation provides information on the type of cable, the conductor material, the insulation and sheath materials, the no. of cores, and the principle design features in abbreviated and simplified form.

The type designation is made up of 18 digits or characters. The type of the conductor is specified first and then the cable construction from inside to outside.

You can order our product either by giving the **alfanar** item code stated in the catalogue or if the required cable construction is not included in our catalogue, you can use the following codes to determine the type of cable you require.

1. Type of conductor material

C :	Copper
A :	Aluminum

2. Type of conductor

1 :	Solid
2 :	Stranded - Circular round
3 :	Stranded - Circular compacted
4 :	Stranded - Sector shaped
5 :	Flexible

3 & 4. Size of conductor

08 :	1.5 mm^2
10 :	2.5 mm^2
12 :	4 mm^2
13 :	6 mm^2
14 :	10 mm^2
15 :	16 mm^2
16 :	25 mm^2
17 :	35 mm^2
18 :	50 mm^2
19 :	70 mm^2
32 :	3 x 10 + 6 mm^2
33 :	3 x 16 + 10 mm^2
34 :	3 x 25 + 16 mm^2
35 :	3 x 35 + 16 mm^2
36 :	3 x 50 + 25 mm^2
37 :	3 x 70 + 35 mm^2
38 :	3 x 95 + 50 mm^2
39 :	3 x 120 + 70 mm^2
40 :	3 x 150 + 70 mm^2
41 :	3 x 185 + 95 mm^2
42 :	3 x 240 + 120 mm^2
43 :	3 x 300 + 150 mm^2
44 :	3 x 400 + 185 mm^2
66 :	3 x 500 + 240 mm^2
45 :	95 mm^2
46 :	120 mm^2
47 :	150 mm^2
48 :	185 mm^2
49 :	240 mm^2
50 :	300 mm^2
51 :	400 mm^2
52 :	500 mm^2
53 :	630 mm^2
54 :	800 mm^2
55 :	1000 mm^2

5. Type of insulation material

X :	XLPE Insulation
P :	PVC Insulation rated 70 ºC
M :	PVC Insulation rated 90 ºC

6. Rated voltage

A :	0.6 / 1 kV

7. Cable construction

1 :	Cores
2 :	Pairs

8 & 9. Number of cores

01 :	1 Core
02 :	2 Cores
03 :	3 Cores
04 :	4 Cores
05 :	5 Cores
≈	≈
61 :	61 Cores

10. Shielding

N :	Copper tape
C :	Copper wires
0 :	Without shielding

11. Armouring

0 :	Without armouring
L :	Pure lead sheathed
N :	Lead alloy sheathed
A :	Aluminum wire armoured
B :	Aluminum tape armoured
G :	Galvanized steel tape armoured
W :	Galvanized steel wire armoured
T :	Non-Galva. steel tape armoured
S :	Lead + G. steel tape armoured
D :	Lead + G. steel wire armoured
X :	Lead + Aluminum wire armoured
R :	Lead + Aluminum tape armoured

12. Outer sheath material

M :	PVC Sheath rated 90 ºC
C :	PVC Sheath rated 80 ºC

13. Outer sheath color

B :	Black
R :	Red
E :	Grey
L :	Blue
N :	Brown
G :	Green
O :	Orange
M :	Green / Yellow

14 & 15. Core identification

51 :	1C - Red
01 :	2C - Red, Black
04 :	3C - Red, Yellow, Blue
08 :	4C - Red, Yellow, Blue, Black
12 :	5C - Red, Yellow, Blue, Black, G/Y
21 :	6C & Above - Black + No.

Note: The mentioned colors are the most common for core identification. However, any other colors for core identification can be used upon a customer's request.

16. Design standard

I :	IEC Standard
B :	BS Standard
C :	Customer request

17. Packing type

M :	Wooden drum
T :	Steel drum

18. Cutting length

S :	250 Meter
F :	500 Meter
R :	1000 Meter

Note: The mentioned cutting lengths are the most common. However, any other cutting lengths can be supplied as per a customer's drum schedule.

Annex K: Conversion Table

Multiply	By	To obtain
Weight-Imperial		
Ounces	28.3495	grams
Pounds(Av)	453.59	grams
Pounds(Av)	0.45359	Kilograms
Tons (short)	907.19	Kilograms
Tons (long)	1016.05	Kilograms
Weight-Metric		
Grams	0.03527	Ounces
Grams	0.002205	Pounds
Kilograms	35.274	Ounces
Kilograms	2.2046	Pounds
kilograms	0.001102	Tons (short)
Kilograms	0.0009842	Tons (long)
Miscellaneous-Imperial		
Pounds per 1000 feet	1.48816	kg/km
Pounds per mile	0.28185	kg/km
Pounds per square inch	0.0007031	kg. per square mm
Pounds per square inch	0.07031	kg. per square cm
Pounds per cubic	27.68	grams per cubic cm
Feet per second	18.288	meters per minute
Feet per second	1.09728	Kilometers per hour
Miles per hour	1.60935	Kilometers per hour
Ohms per 1000 feet	3.28083	Ohms per Kilometer
Ohms per mile	0.62137	Ohms per Kilometer
Decibels per 1000 feet	3.28083	Decibels per kilomter
Decibels per mile	0.62137	Decibels per kilomter
Decibels	0.1153	nepers
Miscellaneous-Metric		
kg/km	0.67197	Pounds per 1000 feet
kg/km	3.54795	Pounds per mile
kg.per square mm	1422.34	Pounds per square inch
kg.per square cm	14.2234	Pounds per square inch
Grams per cubic cm	0.03613	Pounds per cubic inch
Meters per minute	0.05468	Feet per second
Kilometers per hour	0.91134	Feet per second
Kilometer per hour	0.62137	Miles per hour
Ohms per Kilometer	0.3048	Ohms per 1000 feet
Ohms per Kilometer	1.6093	Ohms per mile
Decibels per kilomter	0.3048	Decibels per 1000 feet
Decibels per kilometer	1.6093	Decibels per mile
Temperature		
° Fahrenheit	5/9(°F)-32	°Celsius
°Celsius	9/5(°C)+32	°Fahrenheit
Lenght-Imperial		
Mils	0.001	inches
Mils	0.0254	mm

Multiply	By	To obtain
inches	1000	Mils
inches	25.40	mm
inches	2.54	cm
Feet	30.48	cm
Feet	0.3048	Meters
Feet (thousand of)	0.3048	kilometers
Yards	0.9144	Meters
Miles	1. 6093	kilometers
Length-Metric		
Millimeters	39.37	Mils
Millimeters	0.03937	inches
Centimeters	0.3937	inches
Centimeters	0.032808	Feet
Meters	39.37	inches
Meters	3.2808	Feet
Meters	1.0936	Yards
Kilometers	3280.83	Feet
Kilometers	0.62137	Miles
Area-Imperial		
Square mils	1.2732	Circular mils
Square mils	0.000001	Square inches
Circular mils	0.7854	Square mils
Circular mils	0.0000007854	Square inches
Square mils	0.0005067	Square mm
Square inches	1000000	Square mils
Square inches	1273240	Circular mils
Square inches	645.16	Square mm
Square inches	6.4516	Square cm
Square feet	0.09290	Square meters
Square yards	0.8361	Square meters
Area-Metric		
Square millimeters	1973.52	Circular mils
Square millimeters	0.00155	Square inches
Square centimeters	0.155	Square inches
Square meters	10.7639	Square feet
Square meters	1.19599	Square yards
Volume-Imperial		
Cubic inches	16.38706	Cubic cm
Cubic feet	0.028317	Cubic meters
Gallons	4.54609	Liters
Volume-U.S.		
Quarts (liquid)	0.9463	Liters
Gallons	3.7854	Liters
Volume-Metric		
Cubic cm	0.06102	Cubic inches
Cubic meters	35.3145	Cubic feet
Liters	1.05668	quarts (liquid U.S)
Liters	0.26417	gallons (U.S.)

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